



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	253	Media
TechCrunch AI	8	Media
The Verge AI	7	Media
HackerNews	4	Community
MIT Tech Review	2	Media

VERIFIED INTELLIGENCE — 5 stories

VERI

AgentTrust: A Self-Improving Trust Layer for AI-Agent Actions

ArXiv CS.AI +2 sources

100%

arXiv:2606.08539v1 Announce Type: new Abstract: AI agents increasingly take consequential actions -- shell commands, cloud operations, and arbitrary tool-calls -- so a trust layer must decide, per action, whether to allow, warn, block, or escalate. We argue that the right way to...

<https://arxiv.org/abs/2606.08539>

VERI

Show HN: AI Pair Programmer for Emacs

HackerNews +1 source

90% HN 32

I built a pair programmer, using AI, but now am trying to go slow with new languages and paradigms using this package. I have noticed a decline in my programming skills, and wondered why I couldn't use AI as a pair programmer. Why can it watch what I'm doing over my shoulder and...

<https://github.com/jaketothepast/codetutor>

VERI

OpenAI files for IPO, following Anthropic

The Verge AI +1 source

80%

OpenAI on Monday checked off a preliminary step in the IPO race that it and rival Anthropic have been competing in for the better part of a year: The company announced it has confidentially submitted a Form S-1 with the US Securities and Exchange Commission, following...

<https://www.theverge.com/ai-artificial-intelligence/946335/openai-ipo-s-1-confidential>

VERI

Apple announces Siri AI and its next generation of Apple Intelligence

The Verge AI +1 source

80%

Two years after first revealing its plans for Apple Intelligence and a smarter Siri that never fully materialized, at WWDC, Apple just revealed a new set of AI features and a smarter, more personalized Siri. Apple calls Siri AI an "entirely new version of Siri" and says it's...

<https://www.theverge.com/tech/942416/apple-siri-ai-update-wwdc>

VERI

Amazon is launching AI-generated custom merch

The Verge AI +1 source

80%

Amazon is expanding its print-on-demand features to AI-generated designs created using Alexa for Shopping for products like T-shirts, water bottles, and hoodies. Shoppers can use text prompts to generate images that are then printed onto blanks for sale on Amazon. They can then...

<https://www.theverge.com/news/945905/amazon-alexa-shopping-ai-generated-custom-merch-de...>

CONFIRMED SIGNALS — 263 stories

CONF

Show HN: Command Center, the AI coding env for people who care about quality

HackerNews 50% HN 49

Hi HN! We're Jimmy and Ray. Jimmy is a Thiel Fellow with a Ph. D. from MIT who has worked on programming tools for 15 years; Ray became VP of Sales at a \$2B company when he was 19 and has built side-businesses vibe-coding. Last year, we set to answer the question "If AI can..."

<https://www.cc.dev/>

CONF

Show HN: Inbox-beam - notifications in your inbox without sending email

HackerNews 50% HN 13

No summary — see link for full article.

<https://github.com/toyoshi/inbox-beam>

CONF

Show HN: Same PRD → bootable FastAPI app, zero LLM calls (600-line Python)

HackerNews 50% HN 4

No summary — see link for full article.

<https://github.com/Anioko/spec-driven-development>

CONF

Apple's long-awaited AI Siri overhaul is finally here

TechCrunch AI 50%

The idea behind the new "Siri AI" is to turn the assistant from a voice controlled assistant into an AI companion that can do a lot more.

<https://techcrunch.com/2026/06/08/apples-long-awaited-ai-siri-overhaul-is-finally-here/>

0 CONF

Apple's WWDC AI demos looked more real after \$250M false ad settlement

TechCrunch AI 50%

The vibe of Apple's 2026 WWDC keynote felt like a spouse proudly listing all the honey-do-list items tackled. One subtle example: the many AI demos of someone standing, phone in hand.

<https://techcrunch.com/2026/06/08/apples-wwdc-ai-demos-looked-more-real-after-250m-fals...>

1 CONF

OpenAI files confidentially for IPO, following Anthropic

TechCrunch AI 50%

The filing comes a little more than a week after its main rival, Anthropic, also filed to go public, ramping up the race between the two AI firms.

<https://techcrunch.com/2026/06/08/following-anthropic-openai-files-confidentially-for-ipo/>

2 CONF

Apple bets cheaper AI will woo small developers

TechCrunch AI 50%

As AI experimentation grows more expensive, Apple is waiving cloud API costs for developers with fewer than 2 million first-time App Store downloads.

<https://techcrunch.com/2026/06/08/apple-bets-cheaper-ai-will-woo-small-developers/>

3 CONF

Apple will let you build workflows using AI in its new Shortcuts app

TechCrunch AI 50%

Shortcuts gets an AI upgrade, letting you describe the workflow you want in a prompt.

<https://techcrunch.com/2026/06/08/apple-will-let-you-build-workflows-using-ai-in-its-ne...>

4 CONF

The Download: AI-generated lawsuits and virtual power plants for data centers

MIT Tech Review 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. How courts are coping with a flood of AI-generated lawsuits Most days in her chambers, Judge Maritza Braswell, a federal magistrate judge in...

<https://www.technologyreview.com/2026/06/04/1138408/the-download-ai-lawsuits-virtual-po...>

5 CONF

Apple is using AI to fix Safari's extension problem

The Verge AI 50%

Apple is trying to solve one of Safari's biggest weaknesses with AI. Safari has long lacked the robust library of extensions that its rivals have, mainly due to the stringent development requirements from Apple. But now, Apple is inviting users to essentially vibe-code their own...

<https://www.theverge.com/tech/946345/apple-safari-ai-update-extensions>

6 CONF

NotebookLM's Gemini 3.5 upgrade adds a cloud computer and help finding sources

The Verge AI 50%

Google is rolling out "across the board" updates to NotebookLM. The AI-powered note-taking app now uses Google's upgraded Gemini 3.5 model, which will allow it to respond with "more accurate and reliable information," according to a blog post on Monday. Launched in 2023,...

<https://www.theverge.com/tech/944325/google-notebooklm-ai-gemini-update>

7 CONF

Microsoft's AI chief says superintelligence is near, but won't take your job

The Verge AI 50%

Today I'm talking with Mustafa Suleyman, the CEO of Microsoft AI. And I'm actually going to keep today's intro short — I'm working from my wife's family farm this week, as you'll see in the video, but also this is a real burner of an episode. We covered everything from Mustafa's...

<https://www.theverge.com/podcast/944138/microsoft-ai-ceo-mustafa-suleyman-superintellig...>

8 CONF

InA-Probe: Instruction-Aware Active Probing for Time Series Forecasting with LLMs

ArXiv CS.AI 50%

arXiv:2606.08601v1 Announce Type: new Abstract: Large Language Models (LLMs) have recently demonstrated impressive potential for time series forecasting. However, existing methods predominantly rely on passive modality alignment or static task reprogramming, which often fail to...

<https://arxiv.org/abs/2606.08601>

9 CONF

MBABench: Evaluating LLM Agents on End-to-End Spreadsheet Tasks in Finance

ArXiv CS.AI 50%

arXiv:2605.22664v2 Announce Type: replace Abstract: LLM agents are increasingly expected to carry out end-to-end workflows, producing complete artifacts from high-level user instructions. To meet enterprise needs, frontier AI labs have developed agents that can construct entire...

<https://arxiv.org/abs/2605.22664>

0 CONF

Automatic Extraction of Structured Information from Brain MRI Reports Using an Open-Weight Large Language Model

ArXiv CS.AI 50%

arXiv:2606.07721v1 Announce Type: new Abstract: Objectives: Automatic data extraction from free-text radiology reports enables large-scale research, but few studies assessed the performance of large language models (LLMs) on Dutch neuroradiology reports. Methods: We analyzed 947...

<https://arxiv.org/abs/2606.07721>

1 **CONF****More Bang for the Buck: Improving the Inference of Large Language Models at a Fixed Budget using Reset and Discard (ReD)****ArXiv CS.AI** **50%**

arXiv:2601.21522v2 Announce Type: replace-cross Abstract: The performance of large language models (LLMs) on verifiable tasks is usually measured by pass@k, the probability of answering a question correctly at least once in k trials. At a fixed budget, a more suitable metric is...

<https://arxiv.org/abs/2601.21522>

2 **CONF****Reconstructing and forecasting disease trajectories of patients with Alzheimer's disease using routine data in resource-constrained settings****ArXiv CS.AI** **50%**

arXiv:2606.07798v1 Announce Type: new Abstract: Alzheimer's disease is a progressive neurodegenerative disorder, and its progression varies substantially across patients. Existing work aims to forecast patients' future cognitive state, with minimal focus on reconstructing the...

<https://arxiv.org/abs/2606.07798>

3 **CONF****GlobeAudio: A Multilingual Multicultural Benchmark for Naturalistic Evaluation of Large Audio-Language Models****ArXiv CS.AI** **50%**

arXiv:2606.08194v1 Announce Type: cross Abstract: Large Audio-Language Models (LALMs) integrate audio perception and language understanding within a unified framework, enabling a wide range of real-world applications. Despite recent advances, evaluation for LALMs remains heavily...

<https://arxiv.org/abs/2606.08194>

4 **CONF****PRISM: Recovering Instruction Sets from Language Model Activations****ArXiv CS.AI** **50%**

arXiv:2606.09563v1 Announce Type: new Abstract: As LLMs are deployed as agents, reliable monitoring requires knowing not only what they output, but which instructions are steering their behavior. This is difficult when models infer unintended subgoals, follow contextual cues, or...

<https://arxiv.org/abs/2606.09563>

5 **CONF****Scaling Participation in Modular AI Systems****ArXiv CS.AI** **50%**

arXiv:2606.07812v1 Announce Type: new Abstract: Humanity is a mosaic of multifaceted talents and needs, and any truly intelligent AI must reflect that richness. Yet the LLMs used by all are built by the few -- a centralized market of monolithic AI models structurally ill-suited...

<https://arxiv.org/abs/2606.07812>

6 **CONF****Scaffold Effects on GAIA: A Controlled Comparison****ArXiv CS.AI** **50%**

arXiv:2606.08529v1 Announce Type: new Abstract: Published agent capability scores conflate what a model can do with what its scaffold lets it do, and the magnitude of this elicitation gap is not well characterized under controlled conditions. This study executes a pre-registered...

<https://arxiv.org/abs/2606.08529>

7 CONF

Safety is Contextual, LLM-Judges Are Not: Navigating the Rigid Priors of Evaluators

ArXiv CS.AI 50%

arXiv:2606.07874v1 Announce Type: new Abstract: LLMs-as-judges are the only way to evaluate safety at scale. Despite their importance, LLM-judges themselves are rarely evaluated beyond human agreement in simple, static benchmarks. We therefore investigate two under-explored but...

<https://arxiv.org/abs/2606.07874>

8 CONF

The AI Epistemic Deference Index: A Continuous Measure of Sycophancy

ArXiv CS.AI 50%

arXiv:2606.07897v1 Announce Type: new Abstract: Current AI models frequently exhibit epistemic sycophancy, endorsing claims to agree with a user. Existing evaluations typically measure this either by assessing what it takes to make a model shift a binary endorsement or by...

<https://arxiv.org/abs/2606.07897>

9 CONF

Voting Protocols as Coordination Mechanisms for Role-Constrained Multi-Agent Tutoring Systems

ArXiv CS.AI 50%

arXiv:2606.08030v1 Announce Type: cross Abstract: Agentic tutoring systems introduce a coordination challenge: multiple agents may propose different but reasonable interventions, yet only one response can be delivered to the learner. In this paper, we study how voting protocols...

<https://arxiv.org/abs/2606.08030>

0 CONF

Cherry-pick Override: Unsafe Directional Commitment in LLM Judges under Mixed Evidence

ArXiv CS.AI 50%

arXiv:2606.07834v1 Announce Type: cross Abstract: LLM judges increasingly turn verdicts into system commitments. Under mixed evidence (claims with both supporting and refuting sources) this is unsafe: when the schema exposes CONFLICTING as the authorized non-directional verdict,...

<https://arxiv.org/abs/2606.07834>

1 CONF

NutriMLLM: Multimodal Large Language Models for Dietary Micronutrient Analysis

ArXiv CS.AI 50%

arXiv:2606.08948v1 Announce Type: cross Abstract: Comprehensive estimation of dietary micronutrients from food images could improve clinical nutrition care, but training such models requires large multimodal datasets linking diverse foods to complete nutrient profiles. We first...

<https://arxiv.org/abs/2606.08948>

2 CONF

Zero-Shot Learning in Industrial Scenarios: New Large-Scale Benchmark, Challenges and Baseline

ArXiv CS.AI 50%

arXiv:2606.07965v1 Announce Type: new Abstract: Large Visual Language Models (LVLMs) have achieved remarkable success in vision tasks. However, the significant differences between industrial and natural scenes make applying LVLMs challenging. Existing LVLMs rely on user-provided...

<https://arxiv.org/abs/2606.07965>

3 CONF

CrossVLA: Cross-Paradigm Post-Training and Inference Optimization for Vision-Language-Action Models

ArXiv CS.AI 50%

arXiv:2605.21854v2 Announce Type: replace-cross Abstract: Vision-Language-Action (VLA) models have rapidly converged on a small set of architectural patterns: discrete-token autoregression (e.g. OpenVLA) and continuous-action flow-matching (e.g. pi-0.5). Yet preference alignment...

<https://arxiv.org/abs/2605.21854>

4 CONF

UniQL: Towards Dialect-Universal Benchmarking for Text-to-SQL

ArXiv CS.AI 50%

arXiv:2606.08018v1 Announce Type: new Abstract: Existing text-to-SQL benchmarks are largely centered on SQLite, making it difficult to evaluate whether models can generalize across heterogeneous SQL dialects. However, real-world database systems differ substantially in syntax,...

<https://arxiv.org/abs/2606.08018>

5 CONF

Vision Language Model Helps Private Information De-Identification in Vision Data

ArXiv CS.AI 50%

arXiv:2606.09132v1 Announce Type: new Abstract: Visual Language Models (VLMs) have gained significant popularity due to their remarkable ability. While various methods exist to enhance privacy in text-based applications, privacy risks associated with visual inputs remain largely...

<https://arxiv.org/abs/2606.09132>

6 CONF

When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference

ArXiv CS.AI 50%

arXiv:2606.08098v1 Announce Type: new Abstract: Majority voting over sampled answers is the dominant unsupervised aggregator for multi-sample LLM inference. We show that piping the signals every sample carries into a delegation-based aggregator (Propagational Proxy Voting, PPV)...

<https://arxiv.org/abs/2606.08098>

7 CONF

Constrained Paraphrase Consistency for LLM Hallucination Detection

ArXiv CS.AI 50%

arXiv:2606.08158v1 Announce Type: cross Abstract: Large language models (LLMs) can generate factually inconsistent claims, motivating accurate and scalable hallucination detectors. Prior work largely enlarges training sets via synthesis or new annotations, introducing increasing...

<https://arxiv.org/abs/2606.08158>

8 CONF

Cross-LLM Consistency in Inference: Evidence from Shared Interactions

ArXiv CS.AI 50%

arXiv:2606.08129v1 Announce Type: new Abstract: Large language models (LLMs) differ in architecture, training data, and optimization procedures, yet they may still develop similar internal inference patterns. In this paper, we examine this hypothesis using interaction-based...

<https://arxiv.org/abs/2606.08129>

9 CONF

SAGE: An LLM-driven Self Reflective Agentic Framework for Fraud Detection

ArXiv CS.AI 50%

arXiv:2606.08146v1 Announce Type: new Abstract: Fraud detection in payment, e-commerce, and telecommunications systems requires accuracy at the individual level, robustness under severe class imbalance, and ease of understanding for risk managers. Existing methods fall at least...

<https://arxiv.org/abs/2606.08146>

0 CONF

When No Answer Is Correct: Diagnosing Absent Answer Detection for MLLMs in Video Understanding

ArXiv CS.AI 50%

arXiv:2606.08239v1 Announce Type: new Abstract: Multimodal large language models (MLLMs) have made substantial advancements in video understanding, yet the reliability of their responses remains underexplored. This work presents a diagnostic study of absent answer detection for...

<https://arxiv.org/abs/2606.08239>

1 CONF

Data Agents Under Attack: Vulnerabilities in LLM-Driven Analytical Systems

ArXiv CS.AI 50%

arXiv:2606.08661v1 Announce Type: cross Abstract: Data agents integrate LLM-driven reasoning with relational data access, executable analytical tools, and multi-step workflow orchestration, making them increasingly central to enterprise analytics. This integration introduces new...

<https://arxiv.org/abs/2606.08661>

2 CONF

Ablation-Reversible Heads Don't Transfer: A Stress Test for Mechanistic Role Claims in Transformers

ArXiv CS.AI 50%

arXiv:2606.08292v1 Announce Type: new Abstract: In mechanistic interpretability, attention heads are commonly elevated to role claims (e.g., "this head represents addition") when they are necessary for a behavior, encode it linearly, and recover that behavior when restored after...

<https://arxiv.org/abs/2606.08292>

3 CONF

HARBOR: A Harness Framework for Agentic Robot Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2606.08610v1 Announce Type: cross Abstract: Reinforcement learning (RL) has become a powerful paradigm for robot learning, particularly in sim-to-real settings, but its broader adoption remains limited by the engineering pipeline surrounding the algorithms. Building tasks,...

<https://arxiv.org/abs/2606.08610>

4 CONF

In-Context Reinforcement Learning via Communicative World Models

ArXiv CS.AI 50%

arXiv:2508.06659v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) agents often struggle to generalize to new tasks and contexts without updating their parameters, mainly because their learned representations and policies are overfit to the specifics of their...

<https://arxiv.org/abs/2508.06659>

5 CONF

Integrating Deep Learning Demand Forecasting with Multi-Objective Optimization for Circular Coffee Supply Chains: A Data-Driven Framework for Cost, Emissions, and Freshness Management

ArXiv CS.AI 50%

arXiv:2606.08314v1 Announce Type: new Abstract: The coffee supply chain is one of the most complex agri-food networks, marked by geographically dispersed production, multi-tier coordination, and high sensitivity to quality and freshness. While sustainability and digitalization...

<https://arxiv.org/abs/2606.08314>

6 CONF

Hiding in Plain Floats: Steganographic Carriers for Indirect Prompt and Content Injection

ArXiv CS.AI 50%

arXiv:2606.08403v1 Announce Type: cross Abstract: Text-centered prompt-injection defenses assume that the malicious signal is visible in one of the inspected text views. We study a reproducible LLM01-style indirect prompt/content-injection failure mode where that assumption...

<https://arxiv.org/abs/2606.08403>

7 CONF

What Makes a Desired Graph for Relational Deep Learning?

ArXiv CS.AI 50%

arXiv:2606.08491v1 Announce Type: new Abstract: Relational deep learning (RDL) converts relational databases (RDBs) into heterogeneous graphs, but graphs derived directly from database schemas are often not well suited for how graph neural networks (GNNs) perform relational...

<https://arxiv.org/abs/2606.08491>

8 CONF

Explaining Black-Box Language Models: Learning to Optimize Linguistically-Structured Word Subsets

ArXiv CS.AI 50%

arXiv:2606.08497v1 Announce Type: new Abstract: As deep language models (DLMs) are increasingly deployed in high-stakes domains such as healthcare, understanding their decision rationale becomes paramount for ensuring trust, safety, and accountability. However, achieving this...

<https://arxiv.org/abs/2606.08497>

9 CONF

MatMind: A Structure-Activity Knowledge-Driven Generative Foundation Model for Materials Science

ArXiv CS.AI 50%

arXiv:2606.07712v1 Announce Type: cross Abstract: Progress in AI-driven crystal materials science has so far been carried by narrow architectures purpose-built for individual tasks -- graph neural networks for property prediction, diffusion and flow-matching models for crystal...

<https://arxiv.org/abs/2606.07712>

0 CONF

MatSciBench: Benchmarking the Reasoning Ability of Large Language Models in Materials Science

ArXiv CS.AI 50%

arXiv:2510.12171v2 Announce Type: replace Abstract: Large Language Models have shown strong scientific reasoning ability, but their performance on materials science problems remains less studied. To fill this gap, we introduce MatSciBench, a comprehensive college-level benchmark...

<https://arxiv.org/abs/2510.12171>

1 CONF

Quantitative Promise Theory: Intentionality and Inference in Autonomous Agents

ArXiv CS.AI 50%

arXiv:2606.08552v1 Announce Type: new Abstract: I discuss some quantitative representations of Promise Theory for processes involving autonomous agents. Agent models are common in software systems, machine learning, and biology, for example, but may also apply to physics and...

<https://arxiv.org/abs/2606.08552>

2 CONF

Considerations for an Integrated Detector Design at FCC-ee: A Human-AI Exploration

ArXiv CS.AI 50%

arXiv:2606.07564v1 Announce Type: cross Abstract: This report explores detector design considerations for the Future Circular Collider in its electron-positron mode (FCC-ee) through an extended dialogue between a physicist and an AI assistant. Starting from initial "prejudice"...

<https://arxiv.org/abs/2606.07564>

3 CONF

Extending Ontologies: From Dense Embeddings to Hybrid Quantum-Fuzzy Systems

ArXiv CS.AI 50%

arXiv:2606.08658v1 Announce Type: new Abstract: LLMs have revolutionized knowledge representation and retrieval, but lack the explicit modeling that knowledge ontologies possess. This paper surveys the ways that ontologies and knowledge graphs have been integrated with dense...

<https://arxiv.org/abs/2606.08658>

4 CONF

Artificial Intelligence for Mathematical Reasoning: An Integrated Survey of Language Models, Neuro-symbolic Systems, and Verified Discovery

ArXiv CS.AI 50%

arXiv:2606.08728v1 Announce Type: new Abstract: Mathematical reasoning has long served as a stringent test of machine intelligence; over the past decade, it has moved from a niche problem within NLP to one of the most consequential AI frontiers. This survey provides a unified...

<https://arxiv.org/abs/2606.08728>

5 CONF

Diverse Thinking Schemata Elicit Better Reasoning in Large Language Models

ArXiv CS.AI 50%

arXiv:2606.08974v1 Announce Type: new Abstract: Large reasoning models (LRMs) have attracted increasing attention for their ability to solve complex mathematical problems by generating extended reasoning chains. In this work, we focus on two critical yet underexplored aspects of...

<https://arxiv.org/abs/2606.08974>

6 CONF

Order Matters: Unveiling the Hidden Impact of Macro Placement Sequences via Proxy-Guided LLM Evolution

ArXiv CS.AI 50%

arXiv:2606.08904v1 Announce Type: new Abstract: Macro placement is a fundamental step in modern chip physical design, playing a crucial role in determining the solution quality of high-dimensional combinatorial optimization problems. Despite recent advancements in machine...

<https://arxiv.org/abs/2606.08904>

7 CONF

Baichuan-M4: A Clinical-Grade Medical Agent System for Continuous Care

ArXiv CS.AI 50%

arXiv:2606.08982v1 Announce Type: new Abstract: Baichuan-M4 is Baichuan Intelligence's clinical-grade medical large model, designed for \emph{continuous care} rather than single-turn medical question answering. It is built as a coordinated medical agent system around three...

<https://arxiv.org/abs/2606.08982>

8 CONF

The Token Not Taken: Sampling, State, and the Variability of AI Agent Outputs

ArXiv CS.AI 50%

arXiv:2606.08998v1 Announce Type: new Abstract: Agentic AI systems can behave differently across runs: the same request may produce a different plan, a different tool call, a different code edit, or a different final answer. Such variability arises from several layers that are...

<https://arxiv.org/abs/2606.08998>

9 CONF

LATTEArena: An Evaluation Framework for LLM-powered Tabular Feature Engineering (Extended Version)

ArXiv CS.AI 50%

arXiv:2606.09004v1 Announce Type: new Abstract: Feature engineering remains essential for tabular data analysis, and Large Language Models (LLMs) have emerged as a promising paradigm for automating this process, giving rise to LLM-powered AuTomated Tabular feature Engineering...

<https://arxiv.org/abs/2606.09004>

0 CONF

A Multi-Agent System for IPMSM Design Optimization via an FEA-AI Hybrid Approach

ArXiv CS.AI 50%

arXiv:2606.09037v1 Announce Type: new Abstract: Interior permanent magnet synchronous motor (IPMSM) design requires balancing conflicting objectives and multi-physics constraints, while modern optimization workflows face three bottlenecks: manual problem setup, high finite...

<https://arxiv.org/abs/2606.09037>

1 CONF

Personalization Meets Safety: Mechanisms, Risks, and Mitigations in Personalized LLMs

ArXiv CS.AI 50%

arXiv:2606.09038v1 Announce Type: new Abstract: Large Language Models (LLMs) have enabled increasingly personalized interactions by adapting to users' preferences, contexts, and long-term histories. However, the mechanisms that enable personalization also expand the safety...

<https://arxiv.org/abs/2606.09038>

2 CONF

Developing Distance-Aware Physics-Constrained Probabilistic Frameworks for Industrial Prognostics

ArXiv CS.AI 50%

arXiv:2512.08499v3 Announce Type: replace-cross Abstract: Development of reliable and physically interpretable probabilistic frameworks for industrial prognostics remain nascent, and existing literature is often insensitive as inputs move away from the training manifold. In this...

<https://arxiv.org/abs/2512.08499>

3 CONF

AGENTSERVESIM: A Hardware-aware Simulator for Multi-Turn LLM Agent Serving

ArXiv CS.AI 50%

arXiv:2606.09613v1 Announce Type: cross Abstract: Multi-turn LLM agents interleave model calls with external tool invocations, shifting serving from stateless request processing to stateful program execution. Serving these workloads requires scheduling, KV-cache management, and...

<https://arxiv.org/abs/2606.09613>

4 CONF

ComplexConstraints and Beyond: Expert Rubrics for RLVR

ArXiv CS.AI 50%

arXiv:2606.09118v1 Announce Type: new Abstract: As LLM capabilities advance rapidly, the evaluation methods used to assess them increasingly lag behind. Traditional benchmarks relied on programmatic verification of narrow, surface-level constraints, but real-world instruction...

<https://arxiv.org/abs/2606.09118>

5 CONF

Benchmarking Vision-Language-Action Models on SO-101: Failure and Recovery Analysis

ArXiv CS.AI 50%

arXiv:2606.08881v1 Announce Type: cross Abstract: Vision-Language-Action (VLA) models have demonstrated strong generalization in robotic manipulation, yet existing evaluations are primarily conducted in simulation or on expensive robotic platforms, leaving their robustness on...

<https://arxiv.org/abs/2606.08881>

6 CONF

Leveraging Structural Constraints for Diffusion-based Neural TSP Solvers

ArXiv CS.AI 50%

arXiv:2606.09343v1 Announce Type: new Abstract: Neural combinatorial optimization has recently achieved strong results on the Euclidean Traveling Salesman Problem (TSP) using generative models such as diffusion and consistency models. State-of-the-art approaches like FT2T combine...

<https://arxiv.org/abs/2606.09343>

7 CONF

DSFNet: Learning Dual-Domain Spectral Operators for Multi-Modality Spatio-Temporal Forecasting in Urban Transportation Systems

ArXiv CS.AI 50%

arXiv:2606.07695v1 Announce Type: cross Abstract: Multi-Modality Spatio-Temporal Forecasting (MoSTF) extends traditional spatio-temporal forecasting by incorporating diverse traffic modalities. Despite significant recent strides in spatio-temporal modeling, existing approaches...

<https://arxiv.org/abs/2606.07695>

8 CONF

Correct Looks Better: Pairwise Comparisons Reveal Accuracy Rankings

ArXiv CS.AI 50%

arXiv:2606.09409v1 Announce Type: new Abstract: Pairwise comparisons combined with aggregation methods like Elo have become central to evaluating generative models, yet concerns remain that they reward superficial stylistic cues or display judge biases. In a more positive turn,...

<https://arxiv.org/abs/2606.09409>

9 CONF

EinSort: Sorting is All We Need for Tensorizing LLM

ArXiv CS.AI 50%

arXiv:2606.08565v1 Announce Type: cross Abstract: Tensor networks provide efficient representations for compressing large neural networks. By carefully designing shapes and topologies, they can significantly reduce memory and computational costs. However, identifying implicit...

<https://arxiv.org/abs/2606.08565>

0 CONF

WeaveBench: A Long-Horizon, Real-World Benchmark for Computer-Use Agents with Hybrid Interfaces

ArXiv CS.AI 50%

arXiv:2606.09426v1 Announce Type: new Abstract: Computer-use agents (CUAs) increasingly operate in runtimes that combine visual desktop control, command-line execution, code editing, browsers, and external tools. Existing benchmarks, however, often evaluate these interfaces as...

<https://arxiv.org/abs/2606.09426>

1 CONF

SIFT: Selective-Index For Fast Compute of RAG Prefill by Exploiting Attention Invariance

ArXiv CS.AI 50%

arXiv:2606.09441v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) injects LLM queries with relevant documents to improve response quality. This injection increases prompt length and slows time to first token (TTFT). Unlike standard queries, RAG queries have a...

<https://arxiv.org/abs/2606.09441>

2 CONF

Thinking-Based Non-Thinking: Solving the Reward Hacking Problem in Training Hybrid Reasoning Models via Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2601.04805v2 Announce Type: replace Abstract: Large reasoning models (LRMs) have attracted much attention due to their exceptional performance. However, their performance mainly stems from thinking, a long Chain of Thought (CoT), which significantly increase computational...

<https://arxiv.org/abs/2601.04805>

3 CONF

LLM-Orchestrated Conformance Checking in Stroke Care Without Computer-Interpretable Guidelines

ArXiv CS.AI 50%

arXiv:2606.09489v1 Announce Type: new Abstract: Objective: Conformance checking in healthcare seeks to assess whether patient care pathways adhere to clinical guidelines. However, its practical application often depends on the availability of formal, machine-interpretable...

<https://arxiv.org/abs/2606.09489>

4 CONF

Deterministic Integrity Gates for LLM-Assisted Clinical Manuscript Preparation: An Auditable Biomedical Informatics Architecture

ArXiv CS.AI 50%

arXiv:2606.09500v1 Announce Type: new Abstract: Objective. Large language models (LLMs) increasingly draft clinical research manuscripts, but their fluency can hide fabricated citations, numbers that drift from source tables, and unmet reporting-guideline items. Existing tools...

<https://arxiv.org/abs/2606.09500>

5 CONF

DYCP: Dynamic Context Pruning for Long-Form Dialogue with LLMs

ArXiv CS.AI 50%

arXiv:2601.07994v5 Announce Type: replace-cross Abstract: Large Language Models (LLMs) increasingly operate over long-form dialogues with frequent topic shifts. While recent LLMs support extended context windows, efficient management of dialogue history in practice is needed due...

<https://arxiv.org/abs/2601.07994>

6 CONF

AI Scientists Are Only as Good as Their Evidence: A Stratified Ablation of Proprietary Data and Reasoning Skills in Drug-Asset Valuation

ArXiv CS.AI 50%

arXiv:2606.09556v1 Announce Type: new Abstract: AI Scientist agents are often evaluated as if capability were mainly a function of model quality, prompting, or reasoning scaffolds. We test a different hypothesis in drug-asset valuation: for knowledge-intensive scientific...

<https://arxiv.org/abs/2606.09556>

7 CONF

Self-Explainability in Self-Adaptive and Self-Organising Systems: Status and Research Directions

ArXiv CS.AI 50%

arXiv:2606.09568v1 Announce Type: new Abstract: The growing complexity of self-adaptive and self-organising systems, fuelled by advances in Artificial Intelligence (AI), has made them increasingly difficult to understand and trust. While Explainable AI aims to provide insight...

<https://arxiv.org/abs/2606.09568>

8 CONF

From 0-to-1 to 1-to-N: Reproducible Engineering Evidence for MetaAI Recursive Self-Design

ArXiv CS.AI 50%

arXiv:2606.09663v1 Announce Type: new Abstract: Recursive self-design refers to AI-assisted modification of the mechanisms by which an AI system is built, evaluated, and improved. This paper treats MetaAI not as a mature paradigm, but as a working term for a human-seeded,...

<https://arxiv.org/abs/2606.09663>

9 CONF

SpatialWorld: Benchmarking Interactive Spatial Reasoning of Multimodal Agents in Real-World Tasks

ArXiv CS.AI 50%

arXiv:2606.09669v1 Announce Type: new Abstract: Spatial reasoning is a foundational capability for multimodal large language models (MLLMs) to perceive and operate within the physical world. However, existing benchmarks predominantly rely on passive evaluation (e.g., static VQA)...

<https://arxiv.org/abs/2606.09669>

0 CONF

Correlation Is Not Enough: Embedding Human Metadata for Individual Causal Discovery

ArXiv CS.AI 50%

arXiv:2606.09672v1 Announce Type: new Abstract: Ask a pretrained biomedical language model whether "cortisol 28 ug/dL" and "stock-market volatility" are related, and it returns a cosine similarity of 0.83 on a scale where 1.0 means identical. The two share no mechanism. This is...

<https://arxiv.org/abs/2606.09672>

1 CONF

Beyond Probabilistic Similarity: Structural, Temporal, and Causal Limitations of Retrieval-Augmented Generation in the Legal Domain

ArXiv CS.AI 50%

arXiv:2606.09724v1 Announce Type: new Abstract: Retrieval-Augmented Generation (RAG) has become a standard architectural response to unreliability in legal AI, yet high-profile failures, including fabricated citations submitted to courts and anachronistic legal content presented...

<https://arxiv.org/abs/2606.09724>

2 CONF

AVI-Bench: Toward Human-like Audio-Visual Intelligence of Omni-MLLMs

ArXiv CS.AI 50%

arXiv:2606.07643v1 Announce Type: cross Abstract: Recent advances in Omni-Multimodal Large Language Models (Omni-MLLMs) have enabled strong integration of vision, audio, and language. However, their audio-visual intelligence (AVI) remains insufficiently evaluated due to the lack...

<https://arxiv.org/abs/2606.07643>

3 CONF

BRAIN: Bayesian Reasoning via Active Inference for Agentic and Embodied Intelligence in Mobile Networks

ArXiv CS.AI 50%

arXiv:2602.14033v1 Announce Type: cross Abstract: Future sixth-generation (6G) mobile networks will demand artificial intelligence (AI) agents that are not only autonomous and efficient, but also capable of real-time adaptation in dynamic environments and transparent in their...

<https://arxiv.org/abs/2602.14033>

4 CONF

Blockchain Infrastructure for Intelligent Cyber--Physical--Social Systems:Post-Quantum Security, Interoperability, and Trustworthy Data Economies in the Era of Embodied AI

ArXiv CS.AI 50%

arXiv:2606.06895v1 Announce Type: cross Abstract: The deployment of embodied artificial intelligence via world-model-based robotics presents a transformative opportunity for blockchain infrastructure, establishing urgent demand for trustworthy data provenance,...

<https://arxiv.org/abs/2606.06895>

5 CONF

ABLE: Representing and Mapping LLMs via Attribution-Based Large-model Embedding

ArXiv CS.AI 50%

arXiv:2606.07524v1 Announce Type: cross Abstract: The explosive growth of large language models (LLMs) has created a heterogeneous and poorly documented ecosystem, making systematic model comparison increasingly important for provenance auditing, security analysis, and model...

<https://arxiv.org/abs/2606.07524>

6 CONF

Implicit Causal Graph Construction in Text via Chain Discovery

ArXiv CS.AI 50%

arXiv:2606.07525v1 Announce Type: cross Abstract: Causal graphs in text are typically populated by observable, predefined events. In contrast, we study implicit causal graph construction from text by treating each described cause-effect pair as the begin- and endpoint of an...

<https://arxiv.org/abs/2606.07525>

7 **CONF****Post-training is (Massive) Supervised Learning****ArXiv CS.AI** **50%**

arXiv:2606.07527v1 Announce Type: cross Abstract: The prevailing paradigm for training LLMs has evolved to rely on a massive post-training phase consisting of SFT and RL. In this position paper, we argue that this methodology effectively marks a reversion to the ``pre-train then...

<https://arxiv.org/abs/2606.07527>

8 **CONF****mllm-shap: A Shapley Value Explainability Platform for Text-Audio Multimodal Large Language Models****ArXiv CS.AI** **50%**

arXiv:2606.07531v1 Announce Type: cross Abstract: We introduce mllm-shap, an open-source Python framework designed to extend Shapley Value (SV) explainability from text-only Large Language Models to Multimodal LLMs (MLLMs) processing joint text and audio inputs. While text-based...

<https://arxiv.org/abs/2606.07531>

9 **CONF****Beware of Geeks Bearing Gifts: Building True EU Frontier AI Sovereignty****ArXiv CS.AI** **50%**

arXiv:2606.07536v1 Announce Type: cross Abstract: Frontier artificial intelligence is reshaping all aspects of society, from economic output or military capability to democratic institutions. The EU is entering this transformation from a position of structural dependence:...

<https://arxiv.org/abs/2606.07536>

0 **CONF****Concerns and Strategic Responses of Older Workers Navigating Generative AI in Bridge Employment****ArXiv CS.AI** **50%**

arXiv:2606.07543v1 Announce Type: cross Abstract: Generative AI (GenAI) is transforming workplaces at a rapid pace. This disproportionately affects vulnerable communities, including older workers (OWs) who re-enter the workforce through bridge employment prior to final...

<https://arxiv.org/abs/2606.07543>

1 **CONF****AI-Integrated Learning Management System for Middle School: A Longitudinal Study of Learning Outcomes Through High School and Beyond****ArXiv CS.AI** **50%**

arXiv:2606.07544v1 Announce Type: cross Abstract: Middle school is a key window for building core academic skills and the learning routines students carry into later grades, yet many students still fall behind because help is often limited and comes too late, after they have...

<https://arxiv.org/abs/2606.07544>

2 **CONF****Liberating LLM Capabilities in Full-Duplex Speech Models****ArXiv CS.AI** **50%**

arXiv:2606.07547v1 Announce Type: cross Abstract: Speech-based large language models are typically constrained to spoken replies, which limits their user-facing outputs to what can be verbalized and suppresses text-native capabilities such as code generation, structured...

<https://arxiv.org/abs/2606.07547>

3 CONF

Evaluating Advanced Prompting on Gemini Flash for Multi-Hop Biomedical QA

ArXiv CS.AI 50%

arXiv:2606.07548v1 Announce Type: cross Abstract: The MedHopQA challenge presents a critical test for Large Language Models (LLMs): complex, multi-hop reasoning in the high-stakes biomedical domain. This paper details our direct API-based evaluation of Google's Gemini Flash...

<https://arxiv.org/abs/2606.07548>

4 CONF

Sound and Complete Neurosymbolic Reasoning with LLM-Grounded Interpretations

ArXiv CS.AI 50%

arXiv:2507.09751v3 Announce Type: replace Abstract: Large language models (LLMs) have demonstrated impressive capabilities in natural language understanding and generation, but exhibit problems with logical consistency in their output. How can we harness LLMs' broad-coverage...

<https://arxiv.org/abs/2507.09751>

5 CONF

Selecting New Measurement Locations to Diversify Traffic-Pattern Coverage: A Real-World Evaluation for Total Traffic Volume Estimation

ArXiv CS.AI 50%

arXiv:2606.07556v1 Announce Type: cross Abstract: Accurate measurement of traffic volumes and flows is vital for modern intelligent transportation. However, despite recent technological advances in sensor devices, it is still expensive to install and maintain fixed traffic...

<https://arxiv.org/abs/2606.07556>

6 CONF

Page image classifier fine-tuned on century-spanning archives of scanned documents for further content-specific processing

ArXiv CS.AI 50%

arXiv:2606.07558v1 Announce Type: cross Abstract: Purpose: Digitization projects in the humanities produce vast, heterogeneous archives of historical documents, making manual sorting impractical at scale. This work addresses the need for an automated system to classify scanned...

<https://arxiv.org/abs/2606.07558>

7 CONF

Model Multiplicity for Adversarial Detection in Small Language Model Training on Edge Devices

ArXiv CS.AI 50%

arXiv:2606.07857v1 Announce Type: cross Abstract: The rise of edge-based machine learning has enabled distributed adaptation of language models across mobile and IoT devices, offering privacy preservation and real-time responsiveness. However, distributed fine-tuning of language...

<https://arxiv.org/abs/2606.07857>

8 CONF

Accelerating Birkhoff Projection for Manifold-Constrained Hyper-Connections

ArXiv CS.AI 50%

arXiv:2606.07574v1 Announce Type: cross Abstract: Manifold-constrained hyper-connections (mHCs) have recently been proposed as a principled extension of hyper-connections, where the residual mixing matrices are constrained to be doubly stochastic via projection onto the Birkhoff...

<https://arxiv.org/abs/2606.07574>

9 CONF

Training-Inference Kernel Contracts: Bounding Divergence in Post-Training and Deployment

ArXiv CS.AI 50%

arXiv:2606.07581v1 Announce Type: cross Abstract: A modern post-training pipeline often writes one symbol for its policy, π_θ , while evaluating it through two different programs: a training kernel optimized for autograd and an inference kernel optimized for low-precision,...

<https://arxiv.org/abs/2606.07581>

00 CONF

Multimodal Group Emotion Recognition In-the-Wild Towards a Privacy-Safe Non-Individual Approach

ArXiv CS.AI 50%

arXiv:2606.07585v1 Announce Type: cross Abstract: This thesis addresses group emotion recognition (GER) in-the-wild with a focus on privacy preservation. Unlike traditional emotion recognition methods that rely on individual-level cues such as face, gaze, or voice analysis, this...

<https://arxiv.org/abs/2606.07585>

01 CONF

From Human Guidance to Autonomy: Agent Skill System for End-to-End LLM Deployment on Spatial NPUs

ArXiv CS.AI 50%

arXiv:2606.07586v1 Announce Type: cross Abstract: Spatial neural processing units (NPUs) provide an energy-efficient platform for edge LLM inference, but efficiently deploying an LLM end-to-end on such hardware remains labor-intensive. Although AI coding agents have begun to...

<https://arxiv.org/abs/2606.07586>

02 CONF

SlideCheck: Guiding Self-Supervised Pretraining of Pathology Foundation Models via Dataset Distributions

ArXiv CS.AI 50%

arXiv:2606.07590v1 Announce Type: cross Abstract: Pathology foundation models are pretrained on large streams of WSI-derived patches, while supervision during data construction is often slide-level, sparse, or heterogeneous. This mismatch makes it difficult to understand and...

<https://arxiv.org/abs/2606.07590>

03 CONF

ResearchClawBench: A Benchmark for End-to-End Autonomous Scientific Research

ArXiv CS.AI 50%

arXiv:2606.07591v1 Announce Type: cross Abstract: AI coding agents are increasingly used for scientific work, but their end-to-end autonomous research capability remains difficult to verify. We present ResearchClawBench, a benchmark for evaluating autonomous scientific research...

<https://arxiv.org/abs/2606.07591>

04 CONF

SENTRY: Statistical Reliability Analysis of Vision Transformers Under Soft Errors

ArXiv CS.AI 50%

arXiv:2606.07620v1 Announce Type: cross Abstract: With the growth of Vision Transformers in safety-critical domains like autonomous systems and medical imaging, ensuring their reliability against soft errors is paramount. While ViTs offer state-of-the-art accuracy, their massive...

<https://arxiv.org/abs/2606.07620>

05 CONF

Pharmacogenomic Knowledge Graph Augmentation for Graph Neural Network-Based Drug-Drug Interaction Prediction

ArXiv CS.AI 50%

arXiv:2606.07698v1 Announce Type: cross Abstract: Graph neural networks (GNNs) applied to drug-drug interaction (DDI) prediction rely exclusively on molecular structure encoded as SMILES-derived graphs. Prior work in this series demonstrated that model performance is bounded by...

<https://arxiv.org/abs/2606.07698>

06 CONF

Physics-Guided Sequence-Based Generative Framework for Acoustic Metamaterial Inverse Design

ArXiv CS.AI 50%

arXiv:2606.09266v1 Announce Type: cross Abstract: Acoustic metamaterial (AMM) inverse design is particularly challenging for broadband target responses due to acoustic dispersion: a structure that matches the desired response at one frequency may deviate at others, and modifying...

<https://arxiv.org/abs/2606.09266>

07 CONF

ScaleSweep: Accurate NVFP4 Post-Training Quantization of LLMs via Block Scale Initialization

ArXiv CS.AI 50%

arXiv:2606.07618v1 Announce Type: cross Abstract: NVFP4 is a recently introduced hardware-supported FP4 format that improves the fidelity of 4-bit quantization through fine-grained block scales. However, existing NVFP4 scale initialization methods still primarily rely on AbsMax...

<https://arxiv.org/abs/2606.07618>

08 CONF

Subtitle-Aligned Fine-Tuning of Whisper for Swiss German ASR: Benchmark Contamination, Convention Mismatch, and an Honest Baseline at 25.6% WER (13.8% cWER)

ArXiv CS.AI 50%

arXiv:2606.07608v1 Announce Type: cross Abstract: We present a systematic study of fine-tuning OpenAI's Whisper large-v3 for Swiss German ASR, using 1,367 hours of broadcast speech paired with Standard German subtitles as weak supervision. Through 16 iterative training runs on...

<https://arxiv.org/abs/2606.07608>

09 CONF

MIRAGE: Metadata-Integrated Repository Analysis and Guided Enhancement for MSR Datasets

ArXiv CS.AI 50%

arXiv:2606.07611v1 Announce Type: cross Abstract: This paper proposes an improved approach to the analysis of Mining Software Repositories (MSR) datasets via metadata enrichment, FAIRness assessment, and topic-driven analysis. This research expands upon an earlier dataset...

<https://arxiv.org/abs/2606.07611>

10 CONF

Position: Anthropomorphic Misalignment Research Needs Stronger Evidence

ArXiv CS.AI 50%

arXiv:2606.07612v1 Announce Type: cross Abstract: We argue that many Anthropomorphic Misalignment Research (AMR) studies need stronger evidence to ensure that they can provide a robust foundation for critical safety decisions, such as model deployment and regulation. By...

<https://arxiv.org/abs/2606.07612>

11 CONF

Can You Trust What You See? Human and AI Detection of Synthetic Legal Evidence

ArXiv CS.AI 50%

arXiv:2606.07613v1 Announce Type: cross Abstract: Visual evidence has long been treated as a reliable form of legal proof, but advances in artificial intelligence (AI) are undermining that assumption. This article asks how well humans and frontier multimodal large language...

<https://arxiv.org/abs/2606.07613>

12 CONF

Structured Neuron Pruning in Deep Neural Networks Using Multi-Armed Bandits

ArXiv CS.AI 50%

arXiv:2606.07615v1 Announce Type: cross Abstract: Deep neural networks often contain redundant hidden units. Removing individual weights can reduce parameter count, but unstructured sparsity is not always easy to exploit in standard dense implementations. This paper develops a...

<https://arxiv.org/abs/2606.07615>

13 CONF

HASA: Subnet Allocation for Compute-Constrained Model-Heterogeneous Federated Learning

ArXiv CS.AI 50%

arXiv:2606.07621v1 Announce Type: cross Abstract: Edge services increasingly use federated learning to personalize on-device models while keeping sensitive data local. In practice, deployments must handle heterogeneity in both client resources and local data distributions....

<https://arxiv.org/abs/2606.07621>

14 CONF

Large Language Models Should Learn Personalized Rather Than Aggregated Human Preferences

ArXiv CS.AI 50%

arXiv:2606.07629v1 Announce Type: cross Abstract: Current approaches to aligning large language models (LLMs) aggregate diverse human preferences into a single reward signal, effectively optimizing for a hypothetical "average user" who represents no real person particularly...

<https://arxiv.org/abs/2606.07629>

15 CONF

BCG-FM: A Foundation Model for Ambient Cardiac Health Sensing

ArXiv CS.AI 50%

arXiv:2606.07692v1 Announce Type: cross Abstract: Foundation models for wearable biosignals have matched or exceeded supervised specialists across a range of clinical tasks, yet all rely on modalities that require deliberate user action--wearing a device or visiting a sleep lab....

<https://arxiv.org/abs/2606.07692>

16 CONF

AMN: An Adaptive Multi-Scale Fusion Network with Boundary and Uncertainty Modeling for Nuclei Segmentation

ArXiv CS.AI 50%

arXiv:2606.07633v1 Announce Type: cross Abstract: Accurate classification of nuclei subtypes in histopathology images is critical for downstream tasks including tumor grading, immune infiltrate quantification, and prognosis prediction. Existing approaches rely on either...

<https://arxiv.org/abs/2606.07633>

17 CONF

DOME: Learning Transferable Domain Variables from Sparse Supervision for Test-Time Adaptation

ArXiv CS.AI 50%

arXiv:2606.07646v1 Announce Type: cross Abstract: Test-time adaptation (TTA) aims to align a model to shifting test domains using only unlabeled streaming data. Most existing methods implicitly infer a single global domain distribution, ignoring the multidimensional and...

<https://arxiv.org/abs/2606.07646>

18 CONF

AgroOmni: A Large-Scale Multi-view Agricultural Dataset for Cross-Scale Multimodal Reasoning

ArXiv CS.AI 50%

arXiv:2603.14342v2 Announce Type: replace-cross Abstract: Modern agricultural data is sourced from diverse platforms and spans multiple spatial scales, ranging from ground-level close-up photography to Unmanned Aerial Vehicle (UAV) aerial observation and satellite remote sensing...

<https://arxiv.org/abs/2603.14342>

19 CONF

MM-Matryoshka: Towards Budget-Elastic Visual Document Retrieval via a 2D Multimodal Matryoshka Training Framework

ArXiv CS.AI 50%

arXiv:2606.07654v1 Announce Type: cross Abstract: Multi-vector visual document retrievers achieve strong fine-grained matching by representing each page with multiple vectors from deep Vision-Language Models (VLMs), but this design makes deployment expensive in both storage and...

<https://arxiv.org/abs/2606.07654>

20 CONF

AgentCompile: An LLM-Guided Compiler for Direct CUDA Inference

ArXiv CS.AI 50%

arXiv:2606.07665v1 Announce Type: cross Abstract: Transformer inference increasingly depends on specialized compiler and runtime support, but real model graphs still require semantic decisions about which regions are worth specializing and which CUDA implementation families are...

<https://arxiv.org/abs/2606.07665>

21 CONF

Liquid Neural Networks as a Drop-in Continuous-Time Deformation Field for Dynamic 3D Gaussian Splatting

ArXiv CS.AI 50%

arXiv:2606.07670v1 Announce Type: cross Abstract: Deformable 3D Gaussian Splatting (D-3DGS) re-constructs dynamic scenes from monocular video by deforming a canonical set of 3D Gaussians through a positional-encoded MLP of frame time t . Although fitted to a continuous variable,...

<https://arxiv.org/abs/2606.07670>

22 CONF

DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment

ArXiv CS.AI 50%

arXiv:2606.07678v1 Announce Type: cross Abstract: Safety alignment for large language models relies on preference data, but current pipelines often train on large, redundant datasets. Existing data selection methods typically score each preference pair independently, collapsing...

<https://arxiv.org/abs/2606.07678>

23 CONF

Systematic LLM Translation of Legacy Scientific Code to Differentiable Frameworks: Application to a Land Surface Model

ArXiv CS.AI 50%

arXiv:2606.07681v1 Announce Type: cross Abstract: Differentiable programming offers transformative capabilities for scientific modeling, enabling gradient-based parameter estimation, sensitivity analysis, and data assimilation. Yet, migrating legacy codebases into differentiable...

<https://arxiv.org/abs/2606.07681>

24 CONF

Test-Time Adaptive Composition for Machine Learning as a Service (MLaaS) in IoT Environments

ArXiv CS.AI 50%

arXiv:2606.07685v1 Announce Type: cross Abstract: The dynamic nature of Internet of Things (IoT) environments affects the long-term effectiveness of Machine Learning as a Service (MLaaS) compositions. Existing adaptive composition methods are mainly based on service replacement...

<https://arxiv.org/abs/2606.07685>

25 CONF

Unambiguous Representations in Neural Networks: An Information-Theoretic Approach to Intentionality

ArXiv CS.AI 50%

arXiv:2512.11000v2 Announce Type: replace-cross Abstract: Representations pervade our daily experience, from letters representing sounds to bit strings encoding digital files. While such representations require externally defined decoders to convey meaning, conscious experience...

<https://arxiv.org/abs/2512.11000>

26 CONF

Adaptive Loss Balancing for Noise-Robust GRPO in Generative Recommendation

ArXiv CS.AI 50%

arXiv:2606.08480v1 Announce Type: cross Abstract: Reinforcement learning (RL) presents a promising avenue for enhancing generative recommendation beyond supervised imitation, leveraging reward signals to guide policy improvement. However, its efficacy is critically contingent on...

<https://arxiv.org/abs/2606.08480>

27 CONF

Tianji-Environ: An Autonomous AI Scientist for Atmospheric Environmental Research

ArXiv CS.AI 50%

arXiv:2606.07697v1 Announce Type: cross Abstract: As atmospheric environmental prediction continues to improve, interpretable validation of pollution mechanisms and feedback processes has become a main challenge in atmospheric chemistry. Yet mechanism validation based on complex...

<https://arxiv.org/abs/2606.07697>

28 CONF

EssentialGIN: a new approach for gene essentiality prediction based on graph isomorphism neural networks

ArXiv CS.AI 50%

arXiv:2606.07700v1 Announce Type: cross Abstract: Background: Prediction of essential genes (proteins), is a basic and challenging problem but at the same time very costly and time-consuming in wet-lab experiments. Predicting essential genes, only based on computational methods...

<https://arxiv.org/abs/2606.07700>

29 CONF

FunctionEvolve: Structure-Guided Symbolic Regression with LLMs

ArXiv CS.AI 50%

arXiv:2606.07704v1 Announce Type: cross Abstract: Symbolic regression aims to uncover explicit scientific laws from data. Recent methods use LLMs to guide mutation from background text, which is more directed than random genetic programming. However, exact symbolic recovery...

<https://arxiv.org/abs/2606.07704>

30 CONF

Attention at the Theoretical Minimum: A Mathematics of Arrays Framework for Memory-Optimal Transformer Kernels

ArXiv CS.AI 50%

arXiv:2606.07713v1 Announce Type: cross Abstract: The attention mechanism is the dominant computational bottleneck in modern transformer-based AI. Its standard implementation incurs quadratic memory traffic in the sequence length $\sim n^2$, and DRAM accesses cost $100\text{--}1000\times$...

<https://arxiv.org/abs/2606.07713>

31 CONF

Beyond Point Estimates: Benchmarking Uncertainty Quantification Methods on the AION-1 Astronomical Foundation Model

ArXiv CS.AI 50%

arXiv:2606.07771v1 Announce Type: cross Abstract: Foundation models for astronomical surveys offer powerful learned representations that can be transferred to downstream regression tasks such as galaxy property estimation. However, point predictions alone are insufficient for...

<https://arxiv.org/abs/2606.07771>

32 CONF

SLMJury: Can Small Language Models Judge as Well as Large Ones?

ArXiv CS.AI 50%

arXiv:2606.07810v1 Announce Type: cross Abstract: Large language models (LLMs) are widely used as judges for evaluating model outputs, but their high cost, latency, and opacity limit scalability. We introduce SLMJury, a framework for evaluating small language models (SLMs) as...

<https://arxiv.org/abs/2606.07810>

33 CONF

Jas: AI-Paired Engineering as a Revival of N-Version Programming

ArXiv CS.AI 50%

arXiv:2606.07828v1 Announce Type: cross Abstract: I report a case study in AI-paired software engineering: five working ports of a vector illustration application across Rust, Swift, OCaml, Python, and browser-based platforms, built by a single developer in approximately 120...

<https://arxiv.org/abs/2606.07828>

34 CONF

Beyond Pass/Fail: Using Process Mining to Understand How LLMs Resist (and Fail) Red Team Attacks

ArXiv CS.AI 50%

arXiv:2606.07833v1 Announce Type: cross Abstract: Standard AI red teaming evaluations reduce adversarial campaigns to a single binary outcome, attack success rate (ASR), not taking into account the sequential structure of how models resist or yield to attacks. We propose...

<https://arxiv.org/abs/2606.07833>

35 CONF

Does Persona Make LLMs K-pop Fans? A Pilot Study of LLM-Based Online Concert Audience Agents

ArXiv CS.AI 50%

arXiv:2606.07837v1 Announce Type: cross Abstract: A concert is a collective experience, but recorded performance videos are typically watched alone, stripping away the shared audience presence that makes concerts feel eventful. We investigate whether persona-based LLM audience...

<https://arxiv.org/abs/2606.07837>

36 CONF

Cost-Aware Speculative Execution for LLM-Agent Workflows: An Integrated Five-Dimension Method

ArXiv CS.AI 50%

arXiv:2606.07846v1 Announce Type: cross Abstract: LLM-agent workflows chain model calls and tool invocations, and spend most of their wall-clock time waiting on upstream operations before downstream ones can start. Speculative execution can reclaim that idle time by launching a...

<https://arxiv.org/abs/2606.07846>

37 CONF

The Cross-Architecture Substrate: A Domain-Transcendent, Calibration-Surviving Geometric Invariant of Modern Vision Encoders

ArXiv CS.AI 50%

arXiv:2606.07882v1 Announce Type: cross Abstract: Different vision neural networks -- trained to classify, contrast, reconstruct, or match images to text -- should have correspondingly different internal representations. We report that they do not. After training, the top...

<https://arxiv.org/abs/2606.07882>

38 CONF

Strained Coherence: A Pre-Failure Signal in Coding Agent Execution Trajectories

ArXiv CS.AI 50%

arXiv:2606.07889v1 Announce Type: cross Abstract: LLM-based coding agents sometimes acknowledge a problem in their own reasoning and then proceed anyway. We call this pattern strained coherence: a safety-relevant failure mode in which an agent has information that should change...

<https://arxiv.org/abs/2606.07889>

39 CONF

Decoupling Semantics and Logic: A Training-Free Coarse-to-Fine Pipeline for Video Retrieval-Augmented Generation

ArXiv CS.AI 50%

arXiv:2606.07924v1 Announce Type: cross Abstract: This paper presents our system description for the 2nd Workshop on Multimodal Augmented Generation via Multimodal Retrieval (MAGMaR). Addressing the critical challenges of cross-lingual long-video comprehension, strict persona...

<https://arxiv.org/abs/2606.07924>

40 CONF

Hardening Agent Benchmarks with Adversarial Hacker-Fixer Loops

ArXiv CS.AI 50%

arXiv:2606.08960v1 Announce Type: cross Abstract: Agent benchmarks score submissions with outcome verifiers that are typically hand-written and brittle, leaving them open to reward hacking. We audit 1,968 tasks across five terminal-agent benchmarks and find 323 (16%) hackable by...

<https://arxiv.org/abs/2606.08960>

41 CONF

Reinforcement Learning for Flow-Matching Policies with Density Transport

ArXiv CS.AI 50%

arXiv:2606.08602v1 Announce Type: cross Abstract: We present an online reinforcement learning (RL) algorithm for fine-tuning flow-matching policies in continuous-control problems. Our key insight is to view RL-based policy improvement as a transport of action densities towards...

<https://arxiv.org/abs/2606.08602>

42 CONF

GVC-Seg: Training-Free 3D Instance Segmentation via Geometric Visual Correspondence

ArXiv CS.AI 50%

arXiv:2606.08014v1 Announce Type: cross Abstract: Accurate 3D instance segmentation in point cloud data is critical for machine vision applications. Recent advancements leverage multiple pre-trained foundation models to generate 3D proposals, followed by the application of...

<https://arxiv.org/abs/2606.08014>

43 CONF

IEA: Amateur-Friendly Conversational Image Editing Agent via Three Stages of Multitask Alignment

ArXiv CS.AI 50%

arXiv:2606.08016v1 Announce Type: cross Abstract: Current image editing software often hinges on fixed filters or expert tuning, leaving a gap between amateur users' intent and outcomes. Creations by generative models may contain artifacts, implausible details, or stylistic...

<https://arxiv.org/abs/2606.08016>

44 CONF

Repair Before Veto, When Repair Is Hidden: Quantum-Accessible Features for Repair-Augmented Constraint Learning

ArXiv CS.AI 50%

arXiv:2606.08020v1 Announce Type: cross Abstract: Hard-constraint decision systems usually veto infeasible candidates. This is too rigid when the system can act: if a known affordable repair would make an infeasible candidate feasible and valuable, rejection is a false veto...

<https://arxiv.org/abs/2606.08020>

45 CONF

Semantic Quorum Assurance: Collective Certification for Non-Deterministic AI Infrastructure

ArXiv CS.AI 50%

arXiv:2606.08021v1 Announce Type: cross Abstract: As large language model (LLM) agents are integrated into autonomous cloud operations, distributed systems face a semantic reliability problem: proposer agents can generate production mutations, such as modifying IAM policies,...

<https://arxiv.org/abs/2606.08021>

46 CONF

Sci-Rho: A Multilingual Visually-Grounded Symbolic Benchmark for STEM Problems

ArXiv CS.AI 50%

arXiv:2606.08034v1 Announce Type: cross Abstract: Symbolic benchmarks have emerged as a key approach to assess model robustness under minor modifications to STEM-related questions. However, existing symbolic benchmarks mostly remain limited to mathematical reasoning, lack visual...

<https://arxiv.org/abs/2606.08034>

47 CONF

When Behavioral Safety Evaluation Fails: A Representation-Level Perspective

ArXiv CS.AI 50%

arXiv:2606.08044v1 Announce Type: cross Abstract: Large Language Model (LLM) safety has often been evaluated at the behavior level, which provides limited evidence of internal robustness, as these evaluations target outputs rather than representation-level vulnerability under...

<https://arxiv.org/abs/2606.08044>

48 CONF

"I understand your perspective": LLM Persuasion and Sycophancy through the Lens of Communicative Action Theory

ArXiv CS.AI 50%

arXiv:2606.08076v1 Announce Type: cross Abstract: Large Language Models (LLMs) can generate high-quality arguments, yet their ability to engage in nuanced and persuasive communicative actions remains largely unexplored. This work explores the persuasive potential of LLMs through...

<https://arxiv.org/abs/2606.08076>

49 CONF

Aligned but Not Partner-Specific: Distinguishing How Multimodal LLM Agents Succeed in Reference Games Without Human-Like Conventions

ArXiv CS.AI 50%

arXiv:2606.08081v1 Announce Type: cross Abstract: Repeated reference games test whether interlocutors replace their initially long descriptions with shorter, partner-specific conventions grounded in shared interaction history. Prior work shows that multimodal LLMs fail to become...

<https://arxiv.org/abs/2606.08081>

50 CONF

Fast LLM-Based Semantic Filtering: From a Unified Framework to an Adaptive Two-Phase Method

ArXiv CS.AI 50%

arXiv:2606.08090v1 Announce Type: cross Abstract: Evaluating a natural-language yes/no predicate over a document corpus under an accuracy target - the semantic filter - is a cornerstone of LLM-based data processing. Calling the LLM on every document (the oracle) is prohibitive,...

<https://arxiv.org/abs/2606.08090>

51 CONF

Ego-Pi: VLA Fine-Tuning for Ego-Centric Human and Robot Data

ArXiv CS.AI 50%

arXiv:2606.08107v1 Announce Type: cross Abstract: Robotics faces a fundamental challenge of data scarcity. Unlike language or vision research, there is no internet-scale dataset for robotic manipulation. A promising path forward is to leverage egocentric human data, which can be...

<https://arxiv.org/abs/2606.08107>

52 CONF

Explaining Data Mixing Scaling Laws

ArXiv CS.AI 50%

arXiv:2606.08167v1 Announce Type: cross Abstract: Recent research has established empirical scaling laws to predict model performance on multi-domain data mixtures. However, a theoretical understanding of these model loss behaviors remains absent. In this work, we propose a...

<https://arxiv.org/abs/2606.08167>

53 CONF

The Governance of Human-LLM Interaction: Safety Gating, Civility Steering, and Affective Default Lock-In

ArXiv CS.AI 50%

arXiv:2606.08172v1 Announce Type: cross Abstract: Large language models (LLMs) increasingly mediate high-stakes interactions in finance, medicine, and mental-health support, yet users have limited control over how these systems communicate. We frame interaction style as a...

<https://arxiv.org/abs/2606.08172>

54 CONF

Frequency-Domain Latent Attention Gating for Cross-Domain Token Aggregation

ArXiv CS.AI 50%

arXiv:2606.08191v1 Announce Type: cross Abstract: Token aggregation is a common bottleneck in models that map token representations to sample-level predictions, yet most pooling methods operate only in the original token domain. We propose FLaG, a plug-in aggregation module that...

<https://arxiv.org/abs/2606.08191>

55 CONF

AeroSpectra Sentinel: An Auditable LLM Prompt-Chaining Decision-Support Workflow for Acute Asthma Risk Assessment from Respiratory Sounds and Clinical Signals

ArXiv CS.AI 50%

arXiv:2606.08247v1 Announce Type: cross Abstract: Acute asthma risk assessment requires rapid interpretation of respiratory sounds, oxygenation, airflow limitation, speech ability, work of breathing, mental status, and response to reliever therapy. Conventional audio-only...

<https://arxiv.org/abs/2606.08247>

56 CONF

Contemporary AI lacks the imagination to diverge or negate in science

ArXiv CS.AI 50%

arXiv:2606.08251v1 Announce Type: cross Abstract: Bold projections that artificial intelligence will accelerate scientific discovery have raced ahead of evidence from working scientists, and the field still lacks large-scale, scientist-in-the-loop tests of these claims. Here we...

<https://arxiv.org/abs/2606.08251>

57 CONF

An AI Security Agent for University ACMIS: Multi-Vector Threat Detection and Automated Response

ArXiv CS.AI 50%

arXiv:2606.08270v1 Announce Type: cross Abstract: University Academic Management Information Systems (ACMIS) are high-value targets for a wide spectrum of security threats including brute-force login attacks, payment fraud, privilege escalation, insider data theft, and academic...

<https://arxiv.org/abs/2606.08270>

58 CONF

"So There's a Catch-22 Here": How Early Adopters Who Build Multi-Agent LLM Systems Conceptualize Transparency

ArXiv CS.AI 50%

arXiv:2606.08323v1 Announce Type: cross Abstract: Multi-agent large language model (LLM) systems are rapidly emerging, yet transparency, a cornerstone of responsible AI, remains under-defined in these distributed architectures, which have complexities of inter-agent coordination...

<https://arxiv.org/abs/2606.08323>

59 **CONF****Set-Based Transformer for Atmospheric Compensation in Standoff LWIR Hyperspectral Imaging**

ArXiv CS.AI 50%

arXiv:2606.08324v1 Announce Type: cross Abstract: Passive long-wave infrared (LWIR) hyperspectral imaging under a standoff geometry depends on atmospheric absorption and emission, as well as reflected radiance, thus making atmospheric compensation essential to get knowledge of a...

<https://arxiv.org/abs/2606.08324>

60 **CONF****Agentic Search for Counterfactual Recourse under Fixed LLM Budgets**

ArXiv CS.AI 50%

arXiv:2606.08696v1 Announce Type: cross Abstract: Counterfactual recourse aims to provide actionable feature changes that would alter an unfavorable decision made by a predictive model. In practice, affected individuals often benefit from multiple feasible alternatives rather...

<https://arxiv.org/abs/2606.08696>

61 **CONF****Self-Supervised Vision Transformers for CBCT-Based Detection of Temporomandibular Joint Osteoarthritis**

ArXiv CS.AI 50%

arXiv:2606.08364v1 Announce Type: cross Abstract: Temporomandibular joint osteoarthritis (TMJ OA) is a prevalent degenerative condition whose osseous changes are often subtle on cone-beam CT (CBCT), making automated detection challenging. We study how well the DINO family of...

<https://arxiv.org/abs/2606.08364>

62 **CONF****A large-scale nanocrystal database with aligned synthesis and properties enabling generative inverse design**

ArXiv CS.AI 50%

arXiv:2601.02424v2 Announce Type: replace-cross Abstract: The synthesis of nanocrystals has been highly dependent on trial-and-error, due to the complex correlation between synthesis parameters and physicochemical properties. Although deep learning offers a potential methodology...

<https://arxiv.org/abs/2601.02424>

63 **CONF****Auditing Proprietary Alignment in Large Language Models: A Comparative Framework Without a Ground-Truth Standard**

ArXiv CS.AI 50%

arXiv:2606.08381v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly released and deployed through opaque development and deployment pipelines, enabling model providers to inject intentional, provider-specific policies without officially announcing...

<https://arxiv.org/abs/2606.08381>

64 **CONF****Impacts of Histories and Models on LLM Grading: A Study in Advanced Software Engineering Courses**

ArXiv CS.AI 50%

arXiv:2606.08400v1 Announce Type: cross Abstract: Graduate-level research reading report assessment creates a substantial labor burden for educators. While large language models (LLMs) hold great potential for automating academic grading, their reliability for this specialized...

<https://arxiv.org/abs/2606.08400>

65 CONF

Hacking Generative Perplexity: Why Unconditional Text Evaluation Needs Distributional Metrics

ArXiv CS.AI 50%

arXiv:2606.08417v1 Announce Type: cross Abstract: Diffusion and continuous flow-based language models have emerged as the leading non-autoregressive alternatives to language modeling. Progress in both paradigms is overwhelmingly tracked by generative perplexity (gen-PPL): the...

<https://arxiv.org/abs/2606.08417>

66 CONF

AI Code Sandboxes: A Comparative Security Study. Part 1 of 2 -- Engine-Level Properties (Attack Surface, Leakage, Stackability, CVE History, Patch Cadence, Fuzzing)

ArXiv CS.AI 50%

arXiv:2606.08433v1 Announce Type: cross Abstract: This paper reads six engine-level measurements together -- 1.1 host attack surface, 1.2 information leakage, 1.3 defense-in-depth stackability, 1.4 public CVE history, 1.5 patch cadence, and 1.6 upstream fuzzing posture -- to...

<https://arxiv.org/abs/2606.08433>

67 CONF

Capacity-Controlled Global Attention for Graph Transformers

ArXiv CS.AI 50%

arXiv:2604.17324v2 Announce Type: replace-cross Abstract: Global self-attention drives modern graph transformers, yet the softmax at its core imposes a structural constraint rarely examined directly: every attention row is non-negative and sums to one, so each per-head output is...

<https://arxiv.org/abs/2604.17324>

68 CONF

FlashCP: Load-Balanced Communication-Efficient Context Parallelism for LLM Training

ArXiv CS.AI 50%

arXiv:2606.08476v1 Announce Type: cross Abstract: Context parallelism (CP) is essential for training large-scale, long-context language models, as it partitions sequences to reduce memory overhead. However, existing CP methods suffer from workload imbalance, inefficient kernels,...

<https://arxiv.org/abs/2606.08476>

69 CONF

PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

ArXiv CS.AI 50%

arXiv:2606.08481v1 Announce Type: cross Abstract: Enterprise property graphs vary widely in schema structure, internal terminology, domain assumptions, governance constraints, and user interaction patterns. A deployment-relevant Text2Cypher benchmark therefore reflects the...

<https://arxiv.org/abs/2606.08481>

70 CONF

STELLAR: Spatio-Temporal Environmental Learning with Latent Alignment and Refinement for Long-Tailed Species Distribution Modeling

ArXiv CS.AI 50%

arXiv:2606.08484v1 Announce Type: cross Abstract: Joint Species Distribution Modeling (JSDM) is a key enabler for biodiversity monitoring and conservation planning. However, accurate JSDM faces two coupled challenges: environmental drivers and species distributions are...

<https://arxiv.org/abs/2606.08484>

71 CONF

Tyan-WP: A Wind Power Foundation Model for Ultra-Short-Term Probabilistic Forecasting

ArXiv CS.AI 50%

arXiv:2606.08630v1 Announce Type: cross Abstract: Global wind power capacity, especially in China, is booming, with new farms spanning diverse terrains and climates. The industry urgently needs accurate wind power foundation models to shorten commissioning and accelerate grid...

<https://arxiv.org/abs/2606.08630>

72 CONF

Reconstructing Synthetic SDO/AIA 193 Å EUV Images from He I 10830 Å Observations with Diffusion Model Translator

ArXiv CS.AI 50%

arXiv:2606.08652v1 Announce Type: cross Abstract: Routine full-disk EUV imaging has been available only since the modern era, such as SOHO and SDO. To extend EUV coronal context into earlier periods, we leverage the multi-decade availability of full-disk HeI{ } observations,...

<https://arxiv.org/abs/2606.08652>

73 CONF

FiberTune: Preserving Action-Fiber Visual Residuals in Vision-Language-Action Fine-Tuning

ArXiv CS.AI 50%

arXiv:2606.08653v1 Announce Type: cross Abstract: Action-supervised fine-tuning of vision-language-action (VLA) policies fits demonstrations effectively but constrains only the directions that change predicted actions, leaving visual structure consistent across action-equivalent...

<https://arxiv.org/abs/2606.08653>

74 CONF

Latent Diffusion Policy: Shaping Latent Spaces for Diffusion-Based Robotic Manipulation

ArXiv CS.AI 50%

arXiv:2606.08657v1 Announce Type: cross Abstract: Diffusion-based visuomotor policies operating directly in raw action spaces conflate scene comprehension with trajectory generation within a single denoising process. The resulting velocity field must simultaneously encode scene...

<https://arxiv.org/abs/2606.08657>

75 CONF

Activation Steering Induces Emergent Misalignment: A More Comprehensive Evaluation

ArXiv CS.AI 50%

arXiv:2606.08682v1 Announce Type: cross Abstract: Activation steering has emerged as a popular inference-time technique for modulating the behavior of large language models (LLMs). By constructing a steering vector from examples of a target behavior and injecting it into...

<https://arxiv.org/abs/2606.08682>

76 CONF

Hybrid Neural Network and Conventional Controller Approach for Robust Control of Highly Unstable Systems: Application to Tilt-Rotor Control

ArXiv CS.AI 50%

arXiv:2606.08714v1 Announce Type: cross Abstract: Multirotors are widely used in applications ranging from surveillance to precision agriculture, yet conventional designs remain limited by their under-actuation. Tilt-rotor configurations overcome this limitation by enabling full...

<https://arxiv.org/abs/2606.08714>

77 CONF

APEX4: Efficient Pure W4A4 LLM Inference via Intra-SM Compute Rebalancing

ArXiv CS.AI 50%

arXiv:2606.08761v1 Announce Type: cross Abstract: W4A4 quantization promises full utilization of INT4 Tensor Cores, yet group dequantization overhead on CUDA Cores has driven existing systems to mixed-precision fallbacks. We present the first systematic study of how intra-SM...

<https://arxiv.org/abs/2606.08761>

78 CONF

TeamHerald@CHIPSAL 2026: Hate Speech Detection and Sentiment Analysis of Nepali Memes using Transformer-based Architectures and Ensemble Learning

ArXiv CS.AI 50%

arXiv:2606.08770v1 Announce Type: cross Abstract: The analysis of internet memes in the Nepali language is complicated by frequent code-mixing and a lack of established baseline resources. While memes inherently combine visual and textual elements, this study focuses on a...

<https://arxiv.org/abs/2606.08770>

79 CONF

Evaluating AI Investment Strategies

ArXiv CS.AI 50%

arXiv:2606.08791v1 Announce Type: cross Abstract: We study the problem of auditing a black-box algorithmic decision-maker from observable inputs and outputs alone. Our main result is an exact decomposition: under precisely characterized conditions, the cumulative \emph{regret}...

<https://arxiv.org/abs/2606.08791>

80 CONF

Governance Controls for AI-Generated Test Artifacts in Autonomous Software Testing

ArXiv CS.AI 50%

arXiv:2606.08806v1 Announce Type: cross Abstract: Artificial Intelligence (AI) and Large Language Models (LLMs) are increasingly used in autonomous software testing; however, AI-generated test artifacts often suffer from hallucinations, compliance violations, security risks, and...

<https://arxiv.org/abs/2606.08806>

81 CONF

Knowledge Graphs and Reasoning LLMs for Finding Simple Yet Effective Transcriptomic Perturbation Predictors

ArXiv CS.AI 50%

arXiv:2606.08816v1 Announce Type: cross Abstract: Predicting the effect of an unseen gene knockout perturbation on transcriptomic gene expression remains a highly challenging problem for virtual cell models. Recent progress has been made by leveraging biological knowledge graphs...

<https://arxiv.org/abs/2606.08816>

82 CONF

Intrinsic Selection and Particle Resampling for Inference-Time Scaling Beyond Domain Verifiability

ArXiv CS.AI 50%

arXiv:2606.08850v1 Announce Type: cross Abstract: Inference-Time Scaling (ITS) has largely succeeded in verifiable domains like math and coding, where cheap verification enables scalable output selection. However, extending ITS to tasks prone to systematic failure - driven by...

<https://arxiv.org/abs/2606.08850>

83 CONF

Optimizing Energy-based Neural Network Training with Coherent Ising Machine

ArXiv CS.AI 50%

arXiv:2606.09117v1 Announce Type: cross Abstract: While Ising machines serve as advanced physical solvers for the Ising model, enabling applications in combinatorial optimization and neural network training, their scalability for large-scale neural networks remains constrained by...

<https://arxiv.org/abs/2606.09117>

84 CONF

Few-shot Class-variable Incremental Audio Classification via Prototype Adaptation and Pseudo Class-variable Training

ArXiv CS.AI 50%

arXiv:2606.08898v1 Announce Type: cross Abstract: In the task of few-shot class-incremental audio classification, the number of classes is assumed to always increase without considering the possibility of decrease. However, the number of classes generally increases or decreases...

<https://arxiv.org/abs/2606.08898>

85 CONF

I-Segmenter: Integer-Only Vision Transformer for Efficient Semantic Segmentation

ArXiv CS.AI 50%

arXiv:2509.10334v2 Announce Type: replace-cross Abstract: Vision Transformers (ViTs) have recently achieved strong results in semantic segmentation, yet their deployment on resource-constrained devices remains limited due to their high memory footprint and computational cost....

<https://arxiv.org/abs/2509.10334>

86 CONF

A Universal Dense Football Event Representation Based on TabTransformer

ArXiv CS.AI 50%

arXiv:2606.09327v1 Announce Type: cross Abstract: Football event data constitute a rich spatiotemporal source for quantitative analysis of player actions in team sports. These datasets contain heterogeneous features, combining continuous location coordinates with categorical...

<https://arxiv.org/abs/2606.09327>

87 CONF

Report on CHIIR 2026 Workshop on Generative AI and Academic Search (GAI&AS)

ArXiv CS.AI 50%

arXiv:2606.08936v1 Announce Type: cross Abstract: This report summarizes the CHIIR 2026 Workshop on Generative AI and Academic Search (GAI&AS), which examined how GenAI is reshaping academic search systems and research practices. The workshop brought together researchers in...

<https://arxiv.org/abs/2606.08936>

88 CONF

Sustainability and Artificial Intelligence: Necessary, Challenging, and Promising Intersections

ArXiv CS.AI 50%

arXiv:2606.09006v1 Announce Type: cross Abstract: Both digital economy and digital technology researchers increasingly recognize the need to better address the role that artificial intelligence (AI) plays in shaping the evolution of the environmental, social and governance...

<https://arxiv.org/abs/2606.09006>

89 CONF

Understanding Quantization-Aware Training: Gradients at Quantized Weights Bias to the Low-Loss Basin

ArXiv CS.AI 50%

arXiv:2606.09012v1 Announce Type: cross Abstract: Post-training quantization (PTQ) converts a trained full-precision model into low-bit weights without task-level retraining, while quantization-aware training (QAT) incorporates quantization into the training loop. Although PTQ...

<https://arxiv.org/abs/2606.09012>

90 CONF

Payoff scaling shapes cooperation in LLM agents across languages

ArXiv CS.AI 50%

arXiv:2601.19082v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly deployed as autonomous agents that negotiate, coordinate, and act on behalf of users. Whether they cooperate in such settings is no longer just an academic question, but a central...

<https://arxiv.org/abs/2601.19082>

91 CONF

TRIAGE: Dialectical Reasoning for Explainable Risk Prediction on Irregularly Sampled Medical Time Series with LLMs

ArXiv CS.AI 50%

arXiv:2606.09030v1 Announce Type: cross Abstract: Clinical early warning systems built on electronic health records, in which clinical observations are recorded as irregularly sampled medical time series (ISMTS), must deliver both calibrated risk scores for patient triage and...

<https://arxiv.org/abs/2606.09030>

92 CONF

A Unifying Lens on Reward Uncertainty in RLHF

ArXiv CS.AI 50%

arXiv:2606.09073v1 Announce Type: cross Abstract: Reinforcement learning from human feedback (RLHF) is bottlenecked by \emph{reward hacking}, where the policy exploits errors in a proxy reward model (RM) and produces high RM scores without genuine quality gains. A natural...

<https://arxiv.org/abs/2606.09073>

93 CONF

Context-Aware Deep Learning for Defect Classification in Atomic-Resolution STEM

ArXiv CS.AI 50%

arXiv:2606.09419v1 Announce Type: cross Abstract: Artificial intelligence is rapidly advancing materials characterization, yet most applications in electron microscopy rely solely on image contrast, overlooking the chemical and experimental context that shapes image formation....

<https://arxiv.org/abs/2606.09419>

94 CONF

Addressing Market Regime Changes and Heavy-Tailed Returns in Portfolio Optimization via Bayesian VAR and Elliptical Black-Litterman

ArXiv CS.AI 50%

arXiv:2606.09104v1 Announce Type: cross Abstract: Deep reinforcement learning (DRL) frameworks for portfolio optimization have shown promise for their ability to learn allocation rules dynamically from market data. However, these models fail to account for fat-tailed returns,....

<https://arxiv.org/abs/2606.09104>

95 CONF

Autonomous Incident Resolution at Hyperscale: An Agentic AI Architecture for Network Operations

ArXiv CS.AI 50%

arXiv:2606.09122v1 Announce Type: cross Abstract: Cloud network infrastructure at hyperscale presents unique operational challenges where traditional human-driven incident response cannot keep pace with the volume, velocity, and complexity of failures. This paper presents an...

<https://arxiv.org/abs/2606.09122>

96 CONF

An Enhanced Geometric-Spectral Feature Learning Framework for Airborne Multispectral Point Cloud Classification

ArXiv CS.AI 50%

arXiv:2606.09123v1 Announce Type: cross Abstract: Multispectral point cloud (MPC) is composed of 3D spatial-spectral information, which holds tremendous potential for accurate land-cover classification. However, the representation power of classification models is limited by...

<https://arxiv.org/abs/2606.09123>

97 CONF

From USD Scenes to Knowledge Graphs: Zero-Shot Ontology Grounding with LLMs

ArXiv CS.AI 50%

arXiv:2606.09134v1 Announce Type: cross Abstract: Constructing knowledge graphs from 3D simulation scenes is essential for robot task reasoning, but the key bottleneck, grounding scene objects to formal ontology classes, still relies on manually curated dictionaries that are...

<https://arxiv.org/abs/2606.09134>

98 CONF

Steganography Without Modification: Hidden Communication via LLM Seeds

ArXiv CS.AI 50%

arXiv:2606.09135v1 Announce Type: cross Abstract: We demonstrate that widely deployed Large Language Model (LLM) inference stacks harbor a steganographic channel that requires no modification to model weights, sampling code, or output distributions. The channel exploits a...

<https://arxiv.org/abs/2606.09135>

99 CONF

Crop Recommendation and Agricultural Query Answering System Using Spatio-Temporal Graph Neural Networks and Hybrid Retrieval Augmentation

ArXiv CS.AI 50%

arXiv:2606.09160v1 Announce Type: cross Abstract: This paper presents a unified system designed to support precision agriculture by integrating advanced weather prediction, crop recommendation, and a question-answering tool for farmers. We propose two deep learning models -- a...

<https://arxiv.org/abs/2606.09160>

00 CONF

CANS: Accelerating Multiuser Collaborative Edge Inference via Cooperative Autodidactic NeuroSurgeon

ArXiv CS.AI 50%

arXiv:2606.09175v1 Announce Type: cross Abstract: Recently, mobile edge computing (MEC)-enabled collaborative deep neural network (DNN) inference has emerged as a promising approach for delivering intelligent services to resource-constrained mobile devices. A representative...

<https://arxiv.org/abs/2606.09175>

01 CONF

Pretrained, Frozen, Still Leaking: Auditing Cross-Encoder Attribute Transfer in EEG Foundation Models

ArXiv CS.AI 50%

arXiv:2606.09189v1 Announce Type: cross Abstract: EEG foundation-model releases are usually audited one endpoint at a time: raw-reconstruction, membership inference, identity linkage, or DP-SGD on the downstream head. We audit the same released embeddings under all four...

<https://arxiv.org/abs/2606.09189>

02 CONF

Trustworthy Smart Fabs via Professional Proxies: Scaling Safe and Sustainable by Design (SSbD) through Industrial Data Spaces

ArXiv CS.AI 50%

arXiv:2606.09227v1 Announce Type: cross Abstract: The convergence of the 2026 European Union Safe and Sustainable by Design (SSbD) framework, Corporate Sustainability Due Diligence Directive (CSDDD), and Carbon Border Adjustment Mechanism (CBAM) introduce a severe governance...

<https://arxiv.org/abs/2606.09227>

03 CONF

Conan-embedding-v3: Fusing Modality-Specific Models for Omni-Modal Embedding

ArXiv CS.AI 50%

arXiv:2606.09331v1 Announce Type: cross Abstract: Omni-modal retrieval promises a single embedding space for text, image, video, document, and audio inputs, but building such a unified retriever is difficult since these modalities differ in data distribution, architecture, and...

<https://arxiv.org/abs/2606.09331>

04 CONF

Scaling Neural Network Verification with Tensor Parallelism and Fully Sharded Data Parallelism

ArXiv CS.AI 50%

arXiv:2606.09377v1 Announce Type: cross Abstract: Formal neural network verification -- proving that a network satisfies safety properties for \emph{all} inputs in a specified domain -- is bounded in practice by GPU memory: standard implementations of bound-propagation...

<https://arxiv.org/abs/2606.09377>

05 CONF

SAILS: Surrogate-based Analysis of Interactions via Local Effect Smooths

ArXiv CS.AI 50%

arXiv:2606.09404v1 Announce Type: cross Abstract: Feature interactions drive much of the predictive power of machine learning models, yet existing explanation methods only detect and quantify interactions without revealing their functional form, or visualize only restricted...

<https://arxiv.org/abs/2606.09404>

06 CONF

AI Assurance in UK Defence: Challenges in Operationalising JSP 936

ArXiv CS.AI 50%

arXiv:2606.09414v1 Announce Type: cross Abstract: This report examines practical challenges in operationalising JSP 936 Part 1 for AI assurance in UK Defence. Using a structured interpretive review of the directive's requirements, the analysis identifies eight thematic challenge...

<https://arxiv.org/abs/2606.09414>

07 CONF

Harness Engineering for Physical AI: Robot Middleware Is the Harness Layer

ArXiv CS.AI 50%

arXiv:2606.09416v1 Announce Type: cross Abstract: Robot middleware faces a new role in the era of Physical AI. Learned policies, planners, and vision-language-action (VLA) models now enter deployed robots as causal participants on the control path, but the layer that integrates...

<https://arxiv.org/abs/2606.09416>

08 CONF

Train at Moving Edge: Online-Verified Prompt Selection for Efficient RL Training of Large Reasoning Model

ArXiv CS.AI 50%

arXiv:2603.25184v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) has become essential for post-training large language models (LLMs) in reasoning tasks. While scaling rollouts can stabilize training and enhance performance, the computational overhead is a...

<https://arxiv.org/abs/2603.25184>

09 CONF

A Finetuned SpeechLLM for Joint Multi-Granular L2 Assessment and Natural-Language Rationales

ArXiv CS.AI 50%

arXiv:2606.09470v1 Announce Type: cross Abstract: Automated L2 speech assessment can assign proficiency labels, but often lacks interpretability. We propose a rubric-guided SpeechLLM for multi-aspect, multi-granular assessment, trained with a hybrid objective combining...

<https://arxiv.org/abs/2606.09470>

10 CONF

Memory Beyond Recall: A Dual-Process Cognitive Memory System for Self-Evolving LLM Agents

ArXiv CS.AI 50%

arXiv:2606.09483v1 Announce Type: cross Abstract: Long-term memory for an LLM agent is more than retrieving the right passage at the right time. Current memory systems collapse belief revision, causal coupling, and cross-domain abstraction into a single retrieval surface tuned...

<https://arxiv.org/abs/2606.09483>

11 CONF

Closing the Prior-Posterior Loop: Self-Reflective Molecular Design with Analysis-Driven LLM Iteration

ArXiv CS.AI 50%

arXiv:2606.09520v1 Announce Type: cross Abstract: Can a general-purpose large language model design molecules with the precision of a seasoned chemist? Current LLM-based frameworks answer this question with scalar feedback loops-generate, score, reject-that amount to informed...

<https://arxiv.org/abs/2606.09520>

12 CONF

Model Poisoning Against Federated Model Adaptation with Chain of Bit-Flips

ArXiv CS.AI 50%

arXiv:2606.09548v1 Announce Type: cross Abstract: Federated Learning (FL) allows a set of clients to collectively train a global model without sharing local training data. Giving the responsibility of the training to decentralized actors may lead to poisoning attacks: clients...

<https://arxiv.org/abs/2606.09548>

13 CONF

Shape Formation for the Cooperative Transportation of Arbitrary Objects Using Multi-Agent Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2606.09610v1 Announce Type: cross Abstract: Cooperative object transportation is essential in numerous domains, including industrial to domestic services. A popular transportation strategy is to carry objects on top of multi-robot systems. The corresponding task is...

<https://arxiv.org/abs/2606.09610>

14 CONF

Powering the Future of AI: Navigating the Trade-offs for Europe's Energy Transition and Net-Zero Goals

ArXiv CS.AI 50%

arXiv:2606.09617v1 Announce Type: cross Abstract: The rapid expansion of AI globally has led to the proliferation of energy-intensive hyperscale data centres (DCs), making them as a structurally challenging component in power system planning and operation. Using a spatially...

<https://arxiv.org/abs/2606.09617>

15 CONF

Do Video Foundation Models Understand Intuitive Physics? A Layerwise Probing Analysis

ArXiv CS.AI 50%

arXiv:2606.09646v1 Announce Type: cross Abstract: We study whether pretrained video foundation models encode intuitive-physics information in their frozen representations, and how this information varies across model families, layers, and probe types. Using frozen-feature...

<https://arxiv.org/abs/2606.09646>

16 CONF

An 84-Format Numeric Catalog with Bit-Exact Conformance Vectors: A Vendor-Neutral Reference for FP8, BF16, MXFP4, and Microscaling Formats

ArXiv CS.AI 50%

arXiv:2606.09686v1 Announce Type: cross Abstract: Numeric format proliferation in machine learning hardware -- FP8 (E4M3 and E5M2), BF16, MXFP4, microscaling block formats, and dozens of research variants -- has outpaced the availability of vendor-neutral, bit-exact reference...

<https://arxiv.org/abs/2606.09686>

17 CONF

GenTSE: Enhancing Target Speaker Extraction via a Coarse-to-Fine Generative Language Model

ArXiv CS.AI 50%

arXiv:2512.20978v2 Announce Type: replace-cross Abstract: Language Model (LM)-based generative modeling has emerged as a promising direction for TSE, offering potential for improved generalization and high-fidelity speech. We propose GenTSE, a two-stage decoder-only generative...

<https://arxiv.org/abs/2512.20978>

18 CONF

Data Synthesis and Parameter-Efficient Fine-Tuning for Low-Resource NMT: A Case Study on Q'eqchi' Mayan

ArXiv CS.AI 50%

arXiv:2606.09767v1 Announce Type: cross Abstract: Neural machine translation for digitally low-resource Indigenous languages is often hindered by extreme data scarcity, prompting reliance on extractive web-scraping. To ensure data sovereignty, this study introduces a data...

<https://arxiv.org/abs/2606.09767>

19 CONF

PTL-Diffusion: Manifold-Aware Diffusion with Periodic Terminal Laws

ArXiv CS.AI 50%

arXiv:2606.09816v1 Announce Type: cross Abstract: Standard diffusion models typically use a single time-homogeneous Gaussian terminal distribution as the reference law for generation. While this choice is analytically convenient and empirically powerful, it provides little...

<https://arxiv.org/abs/2606.09816>

20 CONF

IDEQ -- Improving Diffusion Models for the Traveling Salesman Problem (TSP) by Leveraging the Structure of the Solution Space

ArXiv CS.AI 50%

arXiv:2412.13858v2 Announce Type: replace Abstract: We investigate diffusion models to solve the Traveling Salesman Problem. Building on the recent DIFUSCO and T2TCO approaches, we propose IDEQ. IDEQ improves the quality of the solutions by leveraging the constrained structure...

<https://arxiv.org/abs/2412.13858>

21 CONF

HA-VLN 2.0: An Open Benchmark and Leaderboard for Human-Aware Navigation in Discrete and Continuous Environments with Dynamic Multi-Human Interactions

ArXiv CS.AI 50%

arXiv:2503.14229v4 Announce Type: replace Abstract: Vision-and-Language Navigation (VLN) has been studied mainly in either discrete or continuous spaces, with little attention to dynamic, crowded environments. We present HA-VLN 2.0, a unified benchmark introducing explicit...

<https://arxiv.org/abs/2503.14229>

22 CONF

Can Global XAI Methods Reveal Injected Behaviours in LLMs? SHAP vs Rule Extraction vs RuleSHAP

ArXiv CS.AI 50%

arXiv:2505.11189v3 Announce Type: replace Abstract: Large language models (LLMs) can amplify misinformation, undermining societal goals such as the UN SDGs. We study three documented drivers of misinformation (valence framing, information overload, and oversimplification) often...

<https://arxiv.org/abs/2505.11189>

23 CONF

AlphaOPT: Formulating Optimization Programs with Self-Improving LLM Experience Library

ArXiv CS.AI 50%

arXiv:2510.18428v4 Announce Type: replace Abstract: Optimization modeling underlies critical decision-making across industries, yet remains difficult to automate: natural-language problem descriptions must be translated into precise mathematical formulations and executable...

<https://arxiv.org/abs/2510.18428>

24 CONF

2-Step Agent: A Framework for the Interaction of a Decision Maker with AI Decision Support

ArXiv CS.AI 50%

arXiv:2602.21889v3 Announce Type: replace Abstract: Predictions from ML models support human decision making in several fields, including high-stakes ones such as healthcare and the judiciary. Yet, we still lack a clear understanding of how decision makers learn from ML-based...

<https://arxiv.org/abs/2602.21889>

25 CONF

Signals Are Not States: Neuro-Symbolic Safeguards for Culturally Aware Classroom AI

ArXiv CS.AI 50%

arXiv:2603.22793v2 Announce Type: replace Abstract: Classroom AI systems increasingly infer high-level educational states such as engagement, confusion, collaboration, participation, and instructional quality from multimodal and linguistic signals. In multicultural and...

<https://arxiv.org/abs/2603.22793>

26 CONF

Explainable AML Triage with LLMs: Evidence Retrieval and Counterfactual Checks

ArXiv CS.AI 50%

arXiv:2604.19755v2 Announce Type: replace Abstract: Anti-money laundering (AML) transaction monitoring generates large volumes of alerts that must be rapidly triaged by investigators under strict audit and governance constraints. While large language models (LLMs) can summarize...

<https://arxiv.org/abs/2604.19755>

27 CONF

Advancing Mathematics Research with AI-Driven Formal Proof Search

ArXiv CS.AI 50%

arXiv:2605.22763v2 Announce Type: replace Abstract: Large language models (LLMs) increasingly excel at mathematical reasoning, but their unreliability limits their utility in mathematics research. A mitigation is using LLMs to generate formal proofs in languages like Lean. We...

<https://arxiv.org/abs/2605.22763>

28 CONF

LGMT: Logic-Grounded Metamorphic Testing for Evaluating the Reasoning Reliability of LLMs

ArXiv CS.AI 50%

arXiv:2605.23965v2 Announce Type: replace Abstract: Large Language Models (LLMs) achieve strong performance on logical reasoning benchmarks, yet their reliability remains uncertain. Existing evaluations rely on static benchmarks, which fail to assess robustness under logically...

<https://arxiv.org/abs/2605.23965>

29 CONF

WorldCoder-Bench: Benchmarking Physically Grounded 3D World Synthesis

ArXiv CS.AI 50%

arXiv:2606.01869v2 Announce Type: replace Abstract: Large language models (LLMs) are increasingly asked not only to write static interfaces, but to construct executable interactive worlds from natural language. Browser-native 3D, commonly built with Three.js, is a natural next...

<https://arxiv.org/abs/2606.01869>

30 CONF

The Shadow Price of Reasoning: Economic Perspective on Optimal Budget Allocation for LLMs

ArXiv CS.AI 50%

arXiv:2606.03092v2 Announce Type: replace Abstract: Inference-time scaling has emerged as a critical avenue for enhancing Large Language Models' performance, yet real-world deployment is constrained by strict computational budgets. In this work, we formulate inference budget...

<https://arxiv.org/abs/2606.03092>

31 CONF

Strategic Integration of Artificial Intelligence in the C-Suite: The Role of the Chief AI Officer

ArXiv CS.AI 50%

arXiv:2407.10247v3 Announce Type: replace-cross Abstract: The integration of Artificial Intelligence (AI) into corporate strategy has become critical for organizations seeking to maintain competitive advantage in the digital age. Although organizations increasingly rely on AI as...

<https://arxiv.org/abs/2407.10247>

32 CONF

Graph-to-SFILES: Control structure prediction from process topologies using generative artificial intelligence

ArXiv CS.AI 50%

arXiv:2412.00508v2 Announce Type: replace-cross Abstract: Control structure design is an important but tedious step in P&ID development. Generative artificial intelligence (AI) promises to reduce P&ID development time by supporting engineers. Previous research on generative AI...

<https://arxiv.org/abs/2412.00508>

33 CONF

Complement or substitute? How AI increases the demand for human skills

ArXiv CS.AI 50%

arXiv:2412.19754v4 Announce Type: replace-cross Abstract: Artificial Intelligence (AI) is transforming the nature of work, yet there is limited empirical evidence on how it affects demand for human skills. This paper examines whether AI adoption increases the prevalence and...

<https://arxiv.org/abs/2412.19754>

34 CONF

FIT-Print: Towards False-claim-resistant Model Ownership Verification via Targeted Fingerprint

ArXiv CS.AI 50%

arXiv:2501.15509v5 Announce Type: replace-cross Abstract: Model fingerprinting has emerged as a crucial mechanism for safeguarding the intellectual property of open-source models, offering a non-intrusive approach that requires no modifications to the protected model. However,...

<https://arxiv.org/abs/2501.15509>

35 CONF

Robust Renal Mass Segmentation on CT: A Validation Study of an AI-Based Framework

ArXiv CS.AI 50%

arXiv:2505.07573v2 Announce Type: replace-cross Abstract: Renal mass segmentation has important potential to enhance the clinical workflow, especially in settings requiring quantitative assessments. Kidney volume could serve as an important biomarker for renal diseases, with...

<https://arxiv.org/abs/2505.07573>

36 CONF

Harmonia: End-to-End RAG Serving Optimization

ArXiv CS.AI 50%

arXiv:2505.07833v2 Announce Type: replace-cross Abstract: Retrieval-Augmented Generation (RAG) improves the reliability of large language models by integrating external knowledge, but serving RAG pipelines efficiently is challenging because requests traverse heterogeneous...

<https://arxiv.org/abs/2505.07833>

37 CONF

Failure by Interference: Language Models Make Balanced Parentheses Errors When Faulty Mechanisms Overshadow Sound Ones

ArXiv CS.AI 50%

arXiv:2507.00322v2 Announce Type: replace-cross Abstract: Despite remarkable advances in coding capabilities, language models (LMs) still struggle with simple syntactic tasks such as generating balanced parentheses. In this study, we investigate the underlying mechanisms behind...

<https://arxiv.org/abs/2507.00322>

38 CONF

AMix-1: A Pathway to Test-Time Scalable Protein Foundation Model

ArXiv CS.AI 50%

arXiv:2507.08920v4 Announce Type: replace-cross Abstract: We introduce AMix-1, a powerful protein foundation model built on Bayesian Flow Networks and empowered by a systematic training methodology, encompassing pretraining scaling laws, emergent capability analysis, in-context...

<https://arxiv.org/abs/2507.08920>

39 CONF

Learning Task Mixtures from Task Affinities: A Probabilistic Graphical Model for Supervised Fine-Tuning

ArXiv CS.AI 50%

arXiv:2507.12612v4 Announce Type: replace-cross Abstract: Supervised fine-tuning performance for large language models depends strongly on how training budget is distributed across a heterogeneous set of tasks. In practice, mixtures are often fixed using simple heuristics (e.g.,...

<https://arxiv.org/abs/2507.12612>

40 CONF

Decoupling the "What" and "Where" With Polar Coordinate Positional Embeddings

ArXiv CS.AI 50%

arXiv:2509.10534v3 Announce Type: replace-cross Abstract: The attention mechanism in a Transformer architecture matches key to query based on both content -- the what -- and position in a sequence -- the where. We present an analysis indicating that what and where are entangled...

<https://arxiv.org/abs/2509.10534>

41 CONF

Generation Properties of Stochastic Interpolation under Finite Training Set

ArXiv CS.AI 50%

arXiv:2509.21925v2 Announce Type: replace-cross Abstract: This paper investigates the theoretical behavior of generative models under finite training populations. Within the stochastic interpolation generative framework, we derive closed-form expressions for the optimal velocity...

<https://arxiv.org/abs/2509.21925>

42 CONF

XCR-Bench: Benchmarking Cross-Cultural Reasoning in LLMs via Culture-Specific Items and Hall's Triad

ArXiv CS.AI 50%

arXiv:2601.14063v2 Announce Type: replace-cross Abstract: Cross-cultural competence in large language models (LLMs) requires understanding and adapting Culture-Specific Items (CSIs) across varying cultural contexts. However, progress in evaluating this capability remains limited...

<https://arxiv.org/abs/2601.14063>

43 CONF

Projection and Quantisation: A Unifying View of Learning to Hash, from Random Projections to the RAG Era

ArXiv CS.AI 50%

arXiv:2510.04127v2 Announce Type: replace-cross Abstract: Approximate nearest neighbour (ANN) search underpins large-scale retrieval, increasingly within the retrieval-augmented generation pipelines that ground large language models, yet the methods that address it have...

<https://arxiv.org/abs/2510.04127>

44 CONF

Efficient Onboard Vision-Language Inference in UAV-Enabled Low-Altitude Economy Networks via LLM-Enhanced Optimization

ArXiv CS.AI 50%

arXiv:2510.10028v2 Announce Type: replace-cross Abstract: The rapid advancement of Low-Altitude Economy Networks (LAENets) has enabled a variety of applications, including aerial surveillance, environmental sensing, and semantic data collection. To support these scenarios,...

<https://arxiv.org/abs/2510.10028>

45 CONF

SmartMixed: A Two-Phase Training Strategy for Adaptive Activation Function Learning in Neural Networks

ArXiv CS.AI 50%

arXiv:2510.22450v3 Announce Type: replace-cross Abstract: The choice of activation function plays a critical role in neural networks, yet most architectures still rely on fixed, uniform activation functions across all neurons. We introduce SmartMixed, a novel two-phase training...

<https://arxiv.org/abs/2510.22450>

46 CONF

Correcting Mean Bias in Text Embeddings: A Refined Renormalization with Training-Free Improvements on MMTEB

ArXiv CS.AI 50%

arXiv:2511.11041v2 Announce Type: replace-cross Abstract: We find that current sentence-embedding models produce outputs with a consistent bias: every embedding e decomposes as $e = \tilde{e} + \mu$, where the mean μ is near-identical across all sentences. We study two...

<https://arxiv.org/abs/2511.11041>

47 CONF

SMART: Shot-Aware Multimodal Video Moment Retrieval with Audio-Enhanced MLLM

ArXiv CS.AI 50%

arXiv:2511.14143v2 Announce Type: replace-cross Abstract: Video Moment Retrieval is a task in video understanding that aims to localize a specific temporal segment in an untrimmed video based on a natural language query. Despite recent progress in moment retrieval from videos...

<https://arxiv.org/abs/2511.14143>

48 CONF

FADTI: Fourier and Attention Driven Diffusion for Multivariate Time Series Imputation

ArXiv CS.AI 50%

arXiv:2512.15116v2 Announce Type: replace-cross Abstract: Multivariate time series imputation is fundamental in applications such as healthcare, traffic forecasting, and biological modeling, where sensor failures and irregular sampling lead to pervasive missing values. However,...

<https://arxiv.org/abs/2512.15116>

49 CONF

Adversarial Instance Generation and Robust Training for Neural Combinatorial Optimization with Multiple Objectives

ArXiv CS.AI 50%

arXiv:2601.01665v2 Announce Type: replace-cross Abstract: Deep reinforcement learning (DRL) has shown great promise in addressing multi-objective combinatorial optimization problems (MOCOPs). Nevertheless, the robustness of these learning-based solvers has remained...

<https://arxiv.org/abs/2601.01665>

50 CONF

How Context Shapes Truth: Geometric Transformations of Statement-level Truth Representations in LLMs

ArXiv CS.AI 50%

arXiv:2601.06599v2 Announce Type: replace-cross Abstract: Large Language Models (LLMs) often encode whether a statement is true as a vector in their residual stream activations. These vectors, also known as truth vectors, have been studied in prior work, however how they change...

<https://arxiv.org/abs/2601.06599>

51 CONF

Revisiting Training Scale: An Empirical Study of Token Count, Power Consumption, and Parameter Efficiency

ArXiv CS.AI 50%

arXiv:2601.06649v2 Announce Type: replace-cross Abstract: Research in machine learning has questioned whether increases in training token counts reliably produce proportional performance gains in large language models. Building on prior work introducing an energy-aware parameter...

<https://arxiv.org/abs/2601.06649>

52 CONF

MMR-GRPO: Accelerating GRPO-Style Training through Diversity-Aware Reward Reweighting

ArXiv CS.AI 50%

arXiv:2601.09085v2 Announce Type: replace-cross Abstract: Group Relative Policy Optimization (GRPO) has become a standard approach for training mathematical reasoning models; however, its reliance on multiple completions per prompt makes training computationally expensive....

<https://arxiv.org/abs/2601.09085>

53 CONF

A Comparative Study of Student Perspectives on Technical Writing Feedback Quality: Evaluating LLMs, SLMs, and Humans in Computer Science Topics

ArXiv CS.AI 50%

arXiv:2601.11541v2 Announce Type: replace-cross Abstract: To address the scalability of feedback in computer science while mitigating the privacy and cost limitations of commercial Large Language Models (LLMs), this study evaluates a locally hosted Small Language Model (SLM). We...

<https://arxiv.org/abs/2601.11541>

54 CONF

Multimodal Generative Engine Optimization: Rank Manipulation for Vision-Language Model Rankers

ArXiv CS.AI 50%

arXiv:2601.12263v2 Announce Type: replace-cross Abstract: Vision-Language Models (VLMs) integrate visual and textual knowledge into unified representations that increasingly underpin modern retrieval and recommendation systems. However, it remains unclear how reliably these...

<https://arxiv.org/abs/2601.12263>

55 CONF

CURE: Curriculum-guided Multi-task Training for Reliable Anatomy Grounded Report Generation

ArXiv CS.AI 50%

arXiv:2601.15408v2 Announce Type: replace-cross Abstract: Medical vision-language models can automate the generation of radiology reports but struggle with accurate visual grounding and factual consistency. Existing models often misalign textual findings with visual evidence,...

<https://arxiv.org/abs/2601.15408>

56 CONF

Comparative evaluation of training strategies using partially labelled datasets for segmentation of white matter hyperintensities and stroke lesions in FLAIR MRI

ArXiv CS.AI 50%

arXiv:2601.20503v2 Announce Type: replace-cross Abstract: White matter hyperintensities (WMH) and ischaemic stroke lesions (ISL) are key imaging biomarkers of cerebral small vessel disease (SVD) detectable on magnetic resonance imaging (MRI). The development of robust deep...

<https://arxiv.org/abs/2601.20503>

57 CONF

Mechanistic Data Attribution: Tracing the Training Origins of Interpretable LLM Units

ArXiv CS.AI 50%

arXiv:2601.21996v2 Announce Type: replace-cross Abstract: While Mechanistic Interpretability has identified interpretable circuits in LLMs, their causal origins in training data remain elusive. We introduce Mechanistic Data Attribution (MDA), a scalable framework that employs...

<https://arxiv.org/abs/2601.21996>

58 CONF

UA-DCM: Uncertainty-aware Causal Decision Making via Effect Bound Decomposition

ArXiv CS.AI 50%

arXiv:2601.22736v2 Announce Type: replace-cross Abstract: Causal inference from observational data can provide strong evidence for finding the best action in a decision-making scenario without having to perform expensive randomized trials. The causal effect of an action is often...

<https://arxiv.org/abs/2601.22736>

59 CONF

Variational Speculative Decoding: Rethinking Draft Training from Token Likelihood to Sequence Acceptance

ArXiv CS.AI 50%

arXiv:2602.05774v4 Announce Type: replace-cross Abstract: Speculative decoding accelerates inference for (M)LLMs, yet a training-decoding discrepancy persists: while existing methods optimize single greedy trajectories, decoding involves verifying and ranking multiple sampled...

<https://arxiv.org/abs/2602.05774>

60 CONF

Implementing Grassroots Logic Programs with Multiagent Transition Systems and AI (Full Version)

ArXiv CS.AI 50%

arXiv:2602.06934v4 Announce Type: replace-cross Abstract: Grassroots Logic Programs (GLP) is a concurrent logic programming language in which logic variables are partitioned into paired readers and writers. An assignment is produced at most once via a writer and consumed at most...

<https://arxiv.org/abs/2602.06934>

61 CONF

Generative Reasoning Re-ranker

ArXiv CS.AI 50%

arXiv:2602.07774v5 Announce Type: replace-cross Abstract: Recent studies increasingly explore Large Language Models (LLMs) as a new paradigm for recommendation systems due to their scalability and world knowledge. However, existing work has three key limitations: (1) most...

<https://arxiv.org/abs/2602.07774>

62 CONF

CodeTaste: Can LLMs Generate Human-Level Code Refactorings?

ArXiv CS.AI 50%

arXiv:2603.04177v2 Announce Type: replace-cross Abstract: LLM coding agents can generate working code, but their solutions often accumulate complexity, duplication, and architectural debt. Human developers address such issues through refactoring: behavior-preserving program...

<https://arxiv.org/abs/2603.04177>

63 CONF

Component Ablation for Efficient Hybrid Language Model Architectures: Performance, Resilience, and Compression Implications

ArXiv CS.AI 50%

arXiv:2603.22473v2 Announce Type: replace-cross Abstract: Hybrid language models combine softmax attention with linear-time sequence mechanisms such as state-space or linear-attention layers, but the functional contribution of each component type remains insufficiently...

<https://arxiv.org/abs/2603.22473>

64 CONF

Distributional Open-Ended Evaluation of LLM Cultural Value Alignment Based on Value Codebook

ArXiv CS.AI 50%

arXiv:2604.06210v4 Announce Type: replace-cross Abstract: As LLMs are globally deployed, aligning their cultural value orientations is critical for safety and user engagement. However, existing benchmarks face the Construct-Composition-Context ($\$C^3$) challenge: relying on...

<https://arxiv.org/abs/2604.06210>

65 CONF

Quantifying Rodda and Graham Gait Classification from 3D Markerless Kinematics derived from a Single-view Video in a Heterogeneous Pediatric Clinical Cohort

ArXiv CS.AI 50%

arXiv:2605.11314v3 Announce Type: replace-cross Abstract: Cerebral Palsy (CP) is a neurological disorder of movement and the most common cause of lifelong physical disability in childhood. Approximately 75% of children with CP are ambulatory, and accurate gait assessment is...

<https://arxiv.org/abs/2605.11314>

66 CONF

DeltaBox: Scaling Stateful AI Agents with Millisecond-Level Sandbox Checkpoint/Rollback

ArXiv CS.AI 50%

arXiv:2605.22781v2 Announce Type: replace-cross Abstract: LLM-powered AI agents require high-frequency state exploration (e.g., test-time tree search and reinforcement learning), relying on rapid checkpoint and rollback (C/R) of the complete sandbox state, including files and...

<https://arxiv.org/abs/2605.22781>

67 CONF

BORA: Bridging Offline Reinforcement Learning and Online Residual Adaptation for Real-World Dexterous VLA Models

ArXiv CS.AI 50%

arXiv:2605.30226v2 Announce Type: replace-cross Abstract: Vision-Language-Action (VLA) models have emerged as a promising paradigm for grounding visual-language understanding into real-world robotic manipulation. However, dexterous manipulation remains challenging for VLA...

<https://arxiv.org/abs/2605.30226>

68 CONF

Beyond Independent Manipulation: Individual Fairness-aware Strategic Classification with Peer Imitation

ArXiv CS.AI 50%

arXiv:2606.00827v2 Announce Type: replace-cross Abstract: Strategic classification (SC) investigates scenarios where agents manipulate their features to obtain favorable decisions from predictive models. Existing fairness-aware SC approaches primarily focus on group fairness and...

<https://arxiv.org/abs/2606.00827>